

Ninja Factory ERP

Complete System Analysis & ERD Documentation

30 Parts | 15 Modules | 137+ Tables | 90+ Business Rules

16 Departments | 10 ERD Diagrams | 25+ Process Flows

Smart Factory ERP v4.0 | September 2026



NINJA FACTORY ERP — COMPLETE SYSTEM ANALYSIS

Project: Ninja Factory — Enterprise ERP System

Industry: Polymer / Plastics Manufacturing (HDPE, PP Granules, Jerrycans, Caps)

Technology: Vanilla JS SPA + Supabase (PostgreSQL + Auth + Realtime + Storage)

Languages: Arabic (RTL) + English (LTR) — Per-user bilingual

Database: PostgreSQL via Supabase — **170+ tables, 17 departments, 94+ screens**

Security: Centralized RBAC with Row-Level Security (RLS)

PART 1 — COMPLETE SYSTEM ANALYSIS

1.1 System Overview

Ninja Factory ERP is a comprehensive, AI-powered enterprise resource planning system designed for a **polymer manufacturing factory**. The factory produces recycled HDPE/PP plastic granules, industrial jerrycans, and plastic caps/closures.

The system manages the **complete business lifecycle** from raw material procurement through production, quality control, sales, delivery, and financial accounting — alongside full HR, fleet management, and clinic/nursing operations.

1.2 All Actors Identified

#	Actor	Type	Arabic
1	Owner	Human / Administrator	صاحب المصنع
2	HR Manager	Human / Administrator	مدير الموارد البشرية
3	Hall Manager / Department Head	Human User	مدير الصالة
4	Employee	Human User	موظف
5	Accounts Manager / CFO	Human User	المدير المالي
6	Chief Accountant	Human User	كبير المحاسبين
7	Accountant	Human User	محاسب
8	Procurement Manager	Human User	مدير المشتريات
9	Procurement Specialist	Human User	أخصائي مشتريات
10	Warehouse Manager	Human User	مدير المخازن
11	Production Manager	Human User	مدير الإنتاج
12	Quality Manager / QC Inspector	Human User	مدير الجودة
13	Maintenance Manager / Technician	Human User	مدير الصيانة
14	Engineering Manager / Engineer	Human User	المدير الهندسي

#	Actor	Type	Arabic
15	Technical Office	Human User	المكتب الفني
16	Sales Manager	Human User	مدير المبيعات
17	Planning Manager	Human User	مدير التخطيط
18	Logistics Manager	Human User	مدير اللوجستيات
19	IT Admin / IT Support	Human / Administrator	مدير تكنولوجيا المعلومات
20	Driver (Internal)	Human User	سائق داخلي
21	Driver (External)	External Human	سائق خارجي
22	Spare Parts Inspector	Human User	مفتش قطع الغيار
23	Nurse / Nursing Management	Human User	ممرض / إدارة التمريض
24	Supplier (External Portal)	External Human	مورد خارجي
25	Customer / Guest	External Human	عميل / زائر
26	AI Engine	Internal System	محرك الذكاء الاصطناعي
27	Notification Service	Internal System	خدمة الإشعارات
28	Supabase Auth	External System	نظام المصادقة
29	Supabase Realtime	External System	البث المباشر
30	Brevo Email Service	Third-party Service	خدمة البريد الإلكتروني

1.3 All User Roles (from Database Constraints)

hr, employee, owner, hr manager, manager, hall manager,
qc inspector, quality manager, warehouse manager,
maintenance manager, technician, procurement manager,
procurement specialist, purchasing manager, engineering manager,
engineer, technical office, sales manager, planning manager,
accountant, chief accountant, cfo, it support, it manager,
spare parts inspector, supplier_external, nursing management, nurse

1.4 All Departments (16)

#	Department (EN)	Department (AR)
1	Production	الإنتاج
2	Warehouse	المخازن
3	Administration	الإدارة
4	Packaging	التعبئة والتغليف
5	Maintenance	الصيانة
6	Procurement	المشتريات
7	Finance	المالية
8	IT	تكنولوجيا المعلومات
9	Quality	الجودة

#	Department (EN)	Department (AR)
10	Engineering	الهندسة
11	Planning	التخطيط
12	Health & Safety	الصحة والسلامة
13	Secretariat	السكرتارية
14	Sales	المبيعات
15	Logistics	اللوجستيات
16	HR	الموارد البشرية

1.5 All Pages / Screens (94+)

Employee Self-Service Screens

dashboard , hr-personal , my-attendance , scan-checkin , scan-checkout , my-leaves , my-salary , my-overtime , my-loans , my-medical , my-delays , my-missions , my-expenses

HR Management Screens

employees , attendance , leaves , absence-leave , shifts , overtime , all-delays , all-missions , employee-warnings , asset-assignment , payroll , payroll-funding , hr-adjustments , recruitment , hr-ats , documents , performance , uniforms , loans , expenses , complaints , medical-requests , offboarding , training , performance-reviews , hr-qr-generator , friday-work , team-adjustments , nursing-medical-approvals

Communication & Collaboration

announcements , internal-chat , org-directory , shift-swap , calendar , task-management

Administration & Reporting

ai-mind , reports , kpi-dashboard , audit-log , login-history , activity-log-page , document-management , approval-workflows , global-search , screen-permissions , notification-settings , system-settings

Finance Screens

petty-cash , financial-reports , chart-of-accounts , driver-payments

Supply Chain Screens

inventory , purchase-requests , erp-suppliers , supplier-portal , supplier-performance

Sales & Products

erp-sales , erp-products , customer-requests

Operations Screens

nursing-page , engineering , fleet , maintenance , quality , production , spare-parts , logistics , equipment , legal-affairs

1.6 All Features Summary

Module	Feature Count	Key Features
Authentication	12	Login, Logout, Sessions, RBAC, RLS, QR Auth, Multi-company
HR & People	35+	Employees CRUD, Attendance, Leaves, Absence tracking, Shifts, Overtime, Missions, QR check-in/out, Complaints, Disciplinary, Offboarding, Warnings, Performance Reviews, Training, Asset Assignment, Friday Work Requests, Recruitment/ATS
Payroll	15+	Payroll calculation, Salary adjustments, Loans, Medical requests, Late deductions, Expenses
Finance	25+	Treasury, Banks, GL, Journal Entries, Invoices, Payments, Checks, Custodies, Cost Centers, Fixed Assets, Taxes, Financial Closing, Chart of Accounts, Budgets
Procurement	12+	Purchase Requests, Purchase Orders, Quotations, Supplier Management, Supplier Performance (AI), Petty Cash
Inventory	8+	Items, Stock Movements, Transactions, Low Stock Alerts, Categories
Production	10+	Production Orders, BOM, Stages, Batches, Consumption, Traceability
Quality	5+	QC Inspections, Pass/Fail, Defect Tracking, Batch Quality
Sales & CRM	12+	Clients, Sales Orders, Products, Public Catalog, Customer Registration, Quote Requests, Customer Feedback
Fleet & Logistics	18+	Vehicles, Internal/External Drivers, Trips, Incidents, Fleet Maintenance, Fuel, Odometer, Cars, Car Trips
Engineering	6+	Projects, Drawings, Technical Office, Workflow Approvals
Maintenance	10+	Machines, Maintenance Logs, Companies, Visits, Payments, Equipment, Equipment Rentals
Nursing	8+	Clinic Visits, Work Injuries, Rest Permits, Pharmacy, Medication Dispense
Legal	3+	Legal Issues, Case Tracking, Assignment
IT & Admin	20+	IT Tickets, Notifications, Announcements, AI Reports, Global Search, KPIs, Tasks, Calendar, Chat, Documents, Webhooks, Email Templates, Approval Workflows

1.7 All Data Entered by Users

- Employee personal data, attendance check-in/out, leave requests, overtime records
- Mission/errand details, complaints, expense claims, loan applications, medical requests
- Payroll adjustments, salary modifications
- Inventory items, stock movements, purchase requests
- Sales orders, client information, product catalog entries
- Journal entries, invoices, payments, checks
- Engineering project details, technical drawings
- Fleet vehicle data, trip logs, fuel logs, odometer readings
- Clinic visit records, injury reports, medication dispensing
- IT support tickets, legal case details
- KPI definitions, task assignments, calendar events, chat messages

1.8 All Data Generated by System

- Payroll net salary calculations (base + overtime + bonuses - penalties - deductions)
- Auto-generated document numbers (PO-, SO-, INV-, JE-, RCV-, PAY-)
- Late deduction calculations based on delay minutes
- AI scores for recruitment CVs (ATS)

- AI supplier recommendations and performance scoring
- Attendance status (present/absent) based on check-in/out
- Low stock alerts based on min_quantity thresholds
- Fleet alerts (license expiry, insurance expiry, maintenance due)
- QR codes for attendance
- Audit logs and activity logs
- AI-generated reports (monthly/semiannual/annual)
- Search index entries
- Notification records
- Distance calculations (odometer_end - odometer_start)
- Financial closing reports

1.9 All Relationships Between Data

The system has extensive FK relationships connecting: - **users** → ALL employee-related tables (attendance, leaves, payroll, loans, etc.) - **inventory_items** → inventory_transactions, purchase_requests, bom_items, production_consumption - **products** → product_categories, product_images, product_documents, bom, production_batches, sales_orders - **suppliers** → supplier_orders, supplier_transactions, supplier_performance, equipment, purchase_orders - **clients** → sales_orders, finance_invoices, quote_requests - **machines** → maintenance_logs - **fleet_vehicles** → fleet_trips, fleet_incidents, fleet_maintenance - **fleet_internal_drivers** → fleet_trips, fleet_vehicles - **companies** → branches → users (multi-company) - **clinic_visits** → clinic_dispense_logs - **clinic_medications** → clinic_dispense_logs - **approval_workflows** → approval_steps → approval_requests → approval_actions

1.10 All External Services

Service	Purpose
Supabase Auth	User authentication, JWT tokens
Supabase PostgreSQL	Database storage
Supabase Realtime	Live notifications, permission changes
Supabase Storage	File uploads (documents, images, CVs)
Brevo (Sendinblue)	Email delivery service
AI Engine (Custom)	Chatbot, Analytics, Fraud Detection, Recommendations

PART 2 — ACTORS / USERS (DETAILED)

Actor 1: Owner (صاحب المصنع)

Property	Details
Type	Human / Super Administrator
Description	Factory owner with complete system authority. Final decision maker for all critical approvals
Responsibilities	Final approval for leaves, loans, medical disbursements, payroll funding, major purchases, hiring
Permissions	FULL ACCESS to all 94+ screens with all 8 actions (view, create, edit, delete, approve, reject, export, print)

Property	Details
Accessible Sections	ALL 15 modules
Key Actions	Approve/Reject leave requests (final stage), Approve loans, Approve medical disbursements, Release payroll funds, View all dashboards and reports, Configure system settings

Actor 2: HR Manager (مدير الموارد البشرية)

Property	Details
Type	Human / Administrator
Description	Manages all HR operations including employees, attendance, leaves, payroll, recruitment
Responsibilities	Employee lifecycle management, Attendance monitoring, Leave workflow (2nd approval), Payroll processing, Recruitment, Training, Performance reviews, Disciplinary actions
Permissions	Full access to HR, Payroll, Recruitment, Attendance, Training, Performance modules. View access to Nursing
Accessible Sections	HR, Payroll, Communication, Reports, Nursing (view)
Key Actions	CRUD on employees, Process attendance, Approve/Reject leaves (HR stage), Calculate payroll, Manage loans, Process offboarding, Manage recruitment, Issue warnings, Manage training

Actor 3: Hall Manager / Department Head (مدير الصالة)

Property	Details
Type	Human User
Description	Manages a specific department/production hall. First-level approver for employee requests
Responsibilities	Team management, First-level leave approval, Overtime approval, Mission approval, Absence monitoring
Permissions	View/Approve for team's leaves, attendance, overtime. Department-specific views
Accessible Sections	Dashboard, Team management, HR self-service
Key Actions	Approve/Reject leave requests (manager stage), Approve overtime, Approve missions, View team attendance, Submit Friday work requests

Actor 4: Employee (موظف)

Property	Details
Type	Human User
Description	Regular factory worker with self-service access to personal HR data
Responsibilities	Check-in/out attendance, Submit leave requests, Submit expense claims, View own salary/payroll
Permissions	View own data only. Create requests (leave, medical, expense, complaint, mission). View announcements and chat
Accessible Sections	Dashboard, My Info (self-service), Communication, Workplace
Screens	dashboard, hr-personal, my-attendance, scan-checkin, scan-checkout, my-leaves, my-salary, my-overtime, my-loans, my-medical, my-delays, my-missions, my-expenses, complaints, announcements, internal-chat, shift-swap, calendar, task-management

Actor 5: CFO / Chief Accountant / Accountant

Property	Details
Type	Human User
Description	Finance team managing all accounting operations
Responsibilities	General ledger, Treasury management, Invoice processing, Check management, Journal entries, Financial closing, Tax filing, Budget management
Permissions	Full access to Finance module. View access to Payroll, Procurement, Sales
Key Actions	Create/Approve journal entries, Process payments, Manage checks, Financial closing, Tax calculations, Budget allocation, Custody management

Actor 6: Procurement Manager / Specialist

Property	Details
Type	Human User
Description	Manages purchasing, supplier relationships, and procurement workflows
Responsibilities	Purchase request processing, Supplier management, Purchase order creation, Quotation management, Petty cash
Permissions	Full access to Procurement, Suppliers. View access to Inventory, Finance
Key Actions	Approve purchase requests, Create purchase orders, Manage suppliers, Request quotations, Track supplier performance

Actor 7: Warehouse Manager (مدير المخازن)

Property	Details
Type	Human User
Description	Manages inventory, stock levels, and material dispensing
Responsibilities	Inventory management, Stock movement tracking, Material dispensing, Low stock monitoring, Purchase request initiation
Permissions	Full access to Inventory module. View access to Procurement, Production
Key Actions	Add/Update inventory items, Process stock movements (in/out/transfer), Dispense materials to production, Monitor low stock alerts

Actor 8: Production Manager (مدير الإنتاج)

Property	Details
Type	Human User
Description	Manages manufacturing operations, production orders, and batch tracking
Responsibilities	Production order management, BOM management, Stage tracking, Material consumption, Batch management
Permissions	Full access to Production module. View access to Inventory, Quality
Key Actions	Create production orders, Manage BOM, Track manufacturing stages, Record material consumption, Create batches

Actor 9: Quality Manager / QC Inspector

Property	Details
Type	Human User
Description	Quality control for incoming materials and finished products
Responsibilities	QC inspections, Pass/Fail decisions, Defect tracking, Batch quality verification
Permissions	Full access to Quality module. View access to Production, Procurement
Key Actions	Create QC inspections, Pass/Fail materials, Inspect batches, Record defects

Actor 10: Maintenance Manager / Technician

Property	Details
Type	Human User
Description	Manages machine maintenance, facilities, and external maintenance companies
Responsibilities	Machine status monitoring, Maintenance scheduling, External company management, Visit tracking, Payment processing
Permissions	Full access to Maintenance module. View access to Inventory (spare parts)
Key Actions	Log maintenance activities, Update machine status, Manage maintenance companies, Track visits, Process maintenance payments

Actor 11: Engineering Manager / Engineer

Property	Details
Type	Human User
Description	Manages engineering projects and technical drawings
Responsibilities	Project management, Drawing management, Technical specifications, Workflow approvals
Permissions	Full access to Engineering module
Key Actions	Create/manage projects, Upload/approve drawings, Update project progress

Actor 12: Sales Manager (مدير المبيعات)

Property	Details
Type	Human User
Description	Manages sales operations, clients, and product catalog
Responsibilities	Client management, Sales order processing, Product catalog management, Customer registration, Quote handling
Permissions	Full access to Sales, Products, Customer Requests
Key Actions	Create sales orders, Manage clients, Manage product catalog, Process customer registrations, Handle quote requests

Actor 13: Logistics Manager

Property	Details
Type	Human User
Description	Manages fleet, drivers, trips, and transportation
Responsibilities	Fleet management, Driver management, Trip scheduling, Incident tracking
Permissions	Full access to Fleet & Logistics module
Key Actions	Manage vehicles, Assign drivers, Schedule trips, Track incidents, Monitor fuel/odometer

Actor 14: IT Admin / IT Support

Property	Details
Type	Human / System Administrator
Description	System administration, IT support, and permission management
Responsibilities	System settings, Permission management, IT ticket resolution, User management
Permissions	Full access to Admin, Permissions, System Settings, Audit logs
Key Actions	Manage screen permissions, Configure system settings, Resolve IT tickets, View audit/login history

Actor 15-16: Driver (Internal / External)

Property	Internal Driver	External Driver
Type	Human User	External Human
Description	Factory employee driver	External transport company driver
Responsibilities	Drive company vehicles, Record trips	Provide transport services
Key Actions	Record odometer, Log trips, Report incidents	Complete deliveries

Actor 17: Spare Parts Inspector

Property	Details
Type	Human User
Description	Inspects spare parts quality, lifecycle tracking
Responsibilities	Quality inspection of spare parts, Lifecycle assessment, Repair recommendations
Key Actions	Inspect parts, Record damage category, Assess repairability, Recommend disposal/repair

Actor 18: Nurse / Nursing Management

Property	Details
Type	Human User
Description	Factory clinic staff managing employee health

Property	Details
Responsibilities	Clinic visits, Injury recording, Rest permits, Medication dispensing, Pharmacy management
Permissions	Full access to Nursing module. View access to Medical Requests
Key Actions	Record clinic visits, Document injuries, Issue rest permits, Dispense medications, Manage pharmacy stock

Actor 19: Supplier (External Portal)

Property	Details
Type	External Human
Description	External supplier with portal access
Responsibilities	View orders, Track payments, Update delivery status
Permissions	Supplier portal only — view own orders and transactions

Actor 20: Customer / Guest

Property	Details
Type	External Human
Description	Public visitor or potential customer
Responsibilities	Browse product catalog, Submit registration requests, Request quotes
Permissions	Public catalog view, Registration form, Quote request form

Actors 21-24: System Actors

Actor	Type	Purpose
AI Engine	Internal System	Chatbot, Analytics, Fraud Detection, CV Scoring, Supplier Recommendations
Notification Service	Internal System	Push notifications, Email triggers, Real-time alerts
Supabase Auth	External System	JWT authentication, Session management
Brevo Email	Third-party	Email delivery for notifications and reports

PART 3 — SECTIONS / MODULES

Module 1: Authentication & Security

Property	Details
Purpose	Secure user authentication, authorization, session management, and audit trailing
Problem Solved	Ensures only authorized users access specific features; tracks all system activities
Users	ALL users (login), IT Admin (configuration), Owner (full access)

Property	Details
Data Managed	Users, Sessions, Permissions, Login history, Audit logs, Activity logs
Connected To	ALL other modules (every module checks permissions)
Tables	users, login_history, activity_log, audit_log, user_sessions, permissions, role_permissions, screen_permissions, permission_templates, companies, branches, used_qr_codes

Module 2: HR & People Management

Property	Details
Purpose	Complete employee lifecycle management from hiring to offboarding
Problem Solved	Centralized HR operations — attendance tracking, leave management, disciplinary actions, performance evaluation
Users	HR Manager, Hall Manager, Employee, Owner
Data Managed	Employee records, Attendance, Leaves, Overtime, Missions, Complaints, Disciplinary actions, Performance reviews, Training, Asset assignments, Warnings, Absence records, ATS/Recruitment
Connected To	Payroll (salary data), Finance (expense processing), Nursing (medical), IT (tickets)
Tables	users, attendance, leave_requests, overtime, missions, complaints, disciplinary_actions, offboarding, hr_absence_requests, hr_absence_records, hr_violations_warnings, hr_absence_audit, hr_absence_settings, performance_reviews, training_courses, training_enrollments, asset_assignments, employee_warnings, friday_work_requests, ats_applications, late_deductions

Module 3: Payroll & Compensation

Property	Details
Purpose	Calculate and process employee salaries, manage loans and financial benefits
Problem Solved	Automated salary calculation integrating attendance, overtime, deductions, and loan installments
Users	HR Manager, Owner, Employee (view own), Accountant
Data Managed	Payroll records, Loans, Medical requests, Expenses, Late deductions, Salary adjustments
Connected To	HR (attendance/overtime data), Finance (payment processing)
Tables	payroll, loans, medical_requests, expenses, late_deductions, salary_adjustments

Module 4: Finance & Accounting

Property	Details
Purpose	Complete financial management including GL, treasury, invoicing, and tax management
Problem Solved	Double-entry accounting, cash flow tracking, check lifecycle, financial reporting
Users	CFO, Chief Accountant, Accountant, Owner
Data Managed	Safes, Bank accounts, Invoices, Payments, Checks, Custodies, Journal entries, Cost centers, Fixed assets, Taxes, Closings, Budgets
Connected To	Payroll (salary payments), Procurement (purchase invoices), Sales (sales invoices), Maintenance (payments)

Property	Details
Tables	finance_safes, finance_bank_accounts, finance_clients, finance_suppliers, finance_invoices, finance_collections, finance_checks, finance_custodies, finance_journal_entries, finance_journal_lines, finance_cost_centers, finance_fixed_assets, finance_expense_categories, finance_taxes, finance_closings, finance_payments, general_ledger, chart_of_accounts, currencies, exchange_rates, department_budgets, doc_sequences

Module 5: Procurement & Purchasing

Property	Details
Purpose	Manage the complete purchasing lifecycle from request to delivery
Problem Solved	Structured procurement workflow with approvals, supplier management, and spending tracking
Users	Procurement Manager, Procurement Specialist, Warehouse Manager, Owner
Data Managed	Purchase requests, Purchase orders, Suppliers, Supplier orders, Supplier transactions, Supplier performance
Connected To	Inventory (stock replenishment), Finance (invoicing), Quality (incoming inspection)
Tables	purchase_requests, purchase_orders, suppliers, supplier_orders, supplier_transactions, supplier_performance

Module 6: Inventory & Warehouse

Property	Details
Purpose	Track all inventory items, stock levels, and material movements
Problem Solved	Real-time stock visibility, automated low-stock alerts, movement traceability
Users	Warehouse Manager, Production Manager, Procurement Manager
Data Managed	Inventory items, Stock movements, Transactions
Connected To	Procurement (incoming stock), Production (material consumption), Sales (finished goods)
Tables	inventory_items, inventory_transactions, stock_movements

Module 7: Production & Manufacturing

Property	Details
Purpose	Manage manufacturing operations from BOM to finished batch
Problem Solved	Production planning, stage tracking, material consumption, batch traceability
Users	Production Manager, Planning Manager, Quality Manager
Data Managed	Production orders, BOM, Stage logs, Consumption records, Batches
Connected To	Inventory (materials), Quality (QC inspections), Sales (finished products)
Tables	production_orders, bom, bom_items, production_stage_templates, production_stage_logs, production_consumption, production_batches

Module 8: Quality Control

Property	Details
Purpose	Ensure product and material quality through inspections
Problem Solved	Quality gate for incoming materials and finished products
Users	Quality Manager, QC Inspector
Data Managed	QC inspections (incoming materials + finished products)
Connected To	Procurement (incoming material), Production (batch quality), Inventory (accepted/rejected)
Tables	qc_inspections

Module 9: Sales & CRM

Property	Details
Purpose	Manage sales operations, client relationships, and public product catalog
Problem Solved	End-to-end sales cycle, customer self-service catalog, quote management
Users	Sales Manager, Customer/Guest
Data Managed	Clients, Sales orders, Products, Categories, Images, Documents, Customer registrations, Quotes, Feedback
Connected To	Finance (invoicing), Production (production orders), Fleet (delivery), Inventory (stock)
Tables	clients, sales_orders, products, product_categories, product_images, product_documents, customer_registration_requests, quote_requests, customer_feedback

Module 10: Fleet & Logistics

Property	Details
Purpose	Manage company fleet, drivers, trips, and transportation
Problem Solved	Vehicle tracking, trip management, cost tracking, maintenance scheduling
Users	Logistics Manager, Driver (Internal/External), Owner
Data Managed	Vehicles, Drivers, Trips, Incidents, Fleet maintenance, Fuel, Odometer, Cars
Connected To	Sales (delivery), HR (driver employee data), Finance (driver payments)
Tables	fleet_internal_drivers, fleet_external_drivers, fleet_vehicles, fleet_trips, fleet_incidents, fleet_maintenance, cars, car_trips, car_maintenance_logs, logistics_drivers, logistics_movements

Module 11: Engineering

Property	Details
Purpose	Manage engineering projects, technical drawings, and approvals
Problem Solved	Technical project tracking, drawing version control, multi-department approval

Property	Details
Users	Engineering Manager, Engineer, Technical Office
Data Managed	Projects, Drawings, Workflow approvals
Connected To	Maintenance (equipment), Production (production lines)
Tables	engineering_projects, engineering_drawings

Module 12: Maintenance & Facilities

Property	Details
Purpose	Machine maintenance, external company management, equipment tracking
Problem Solved	Preventive maintenance scheduling, repair tracking, equipment rental management
Users	Maintenance Manager, Technician, Owner
Data Managed	Machines, Maintenance logs, Companies, Visits, Payments, Equipment, Rentals
Connected To	Inventory (spare parts), Finance (payments), Engineering (equipment)
Tables	machines, maintenance_logs, maintenance_requests, maintenance_companies, maintenance_visits, maintenance_payments, equipment_categories, equipment, equipment_rentals

Module 13: Nursing & Factory Clinic

Property	Details
Purpose	Factory clinic operations — employee health, injuries, medication
Problem Solved	Occupational health tracking, injury documentation, medication inventory
Users	Nurse, Nursing Management, HR Manager
Data Managed	Clinic visits, Work injuries, Rest permits, Medications, Dispense logs
Connected To	HR (employee data, leave integration), Payroll (medical requests)
Tables	clinic_visits, clinic_injuries, clinic_rest_permits, clinic_medications, clinic_dispense_logs

Module 14: Legal & Compliance

Property	Details
Purpose	Track legal issues, cases, and compliance matters
Problem Solved	Centralized legal case management
Users	Owner, Legal Advisor (assigned_to)
Data Managed	Legal issues
Connected To	HR (employee-related cases)
Tables	legal_issues

Module 15: IT & Administration

Property	Details
Purpose	IT support, system administration, collaboration tools, AI analytics
Problem Solved	IT ticket management, system configuration, internal communication, AI-powered insights
Users	IT Admin, All users (notifications, chat, tasks)
Data Managed	IT tickets, Notifications, Announcements, AI reports, KPIs, Tasks, Calendar, Chat, Documents, Approval workflows, System settings
Connected To	ALL modules (notifications, documents, approvals span entire system)
Tables	it_tickets, notifications, announcements, notification_preferences, system_settings, ai_reports, kpi_definitions, kpi_values, saved_reports, tasks, calendar_events, chat_channels, chat_messages, documents, document_categories, digital_signatures, webhooks, email_templates, import_logs, backup_logs, search_index, approval_workflows, approval_steps, approval_requests, approval_actions

Cross-Module Dependency Map

```
flowchart TD
    AUTH[🔒 Authentication & Security] --> HR[👤 HR & People]
    AUTH --> FIN[💰 Finance]
    AUTH --> ALL[All Modules]

    HR --> PAY[💰 Payroll]
    HR --> NURSE[🏥 Nursing]
    HR --> FLEET[🚚 Fleet - Drivers]

    PAY --> FIN

    PROC[📦 Procurement] --> INV[📦 Inventory]
    PROC --> FIN
    PROC --> QC[🔍 Quality]

    INV --> PROD[🏭 Production]
    PROD --> QC
    PROD --> SALES[📊 Sales]

    SALES --> FIN
    SALES --> FLEET

    MAINT[🔧 Maintenance] --> INV
    MAINT --> FIN

    ENG[⚙️ Engineering] --> PROD
    ENG --> MAINT

    FLEET --> FIN

    IT[💻 IT & Admin] --> ALL
    LEGAL[⚖️ Legal] --> HR
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PART 4 — SECTION 1: Authentication & Security

4.1 Section Purpose

The Authentication & Security module is the **foundation of the entire ERP system**. It controls:

- **Who** can access the system (Authentication)
- **What** they can see and do (Authorization / RBAC)
- **When** and **how** they accessed it (Audit Trail)
- **Multi-company** isolation (Company/Branch scoping)

Why it exists: Without this module, sensitive factory data (salaries, financial records, production secrets) would be unprotected.

What problem it solves: Centralized Role-Based Access Control ensuring that a Production worker cannot see payroll data, and only authorized managers can approve requests.

Which users interact: ALL users must authenticate. IT Admin and Owner configure permissions.

Data managed: User accounts, sessions, permissions, login history, activity logs, audit trails, QR codes.

4.2 Actors

Actor	Can Do	Cannot Do	Data Access	Processes Initiated	Processes Received
Owner	Full system access, configure permissions, view all audit logs	N/A (unrestricted)	All tables	Permission changes, User creation	Audit alerts
HR Manager	Create/Edit employees, assign roles, view login history	Change system-level settings	users, login_history, activity_log	User creation, Role assignment	Permission notifications
IT Admin	Manage screen_permissions, system_settings, view audit logs	Approve HR/Finance requests	All security tables	Permission configuration, System settings	IT tickets
Employee	Login, Logout, Change own password, QR check-in/out	View other users' data, change permissions	Own user record only	Login, QR check-in	Notifications
Supabase Auth	Validate JWT tokens, Manage sessions	N/A (system)	user_sessions	Token validation	Auth requests
AI Engine	Query data within permission scope	Access data outside user's permission	Filtered by AIPermissionLayer	Analytics queries	Chatbot requests

4.3 Features

Feature 1: User Login

Property	Details
Description	User authenticates with username + password to receive session token

Property	Details
Actor	Any User
Input	username, password
Output	Session token, User profile, Permissions list
Tables	users, login_history, user_sessions
CRUD	R (users), C (login_history, user_sessions)
Business Rules	BR-AUTH-01: Username and password must match. BR-AUTH-02: User status must be 'active'. BR-AUTH-03: Failed login must be recorded with reason
Permissions	Public (no permission needed to access login page)
Errors	Invalid credentials, Account suspended, Account inactive, Account locked

Feature 2: User Logout

Property	Details
Description	Terminate active session and clear client tokens
Actor	Authenticated User
Input	Session token
Output	Session invalidated
Tables	user_sessions
CRUD	U (user_sessions.is_active = false)
Business Rules	BR-AUTH-04: Session must be deactivated on logout

Feature 3: Session Management

Property	Details
Description	Track active sessions with token hashing and expiry
Actor	System (automatic)
Input	User login event
Output	Active session record
Tables	user_sessions
CRUD	C (on login), R (on validation), U (on refresh), D (on expiry/logout)
Business Rules	BR-AUTH-05: Tokens stored as hashes only. BR-AUTH-06: Sessions expire after configured time. BR-AUTH-07: Refresh tokens extend session

Feature 4: Screen-Level Permission Check (canView)

Property	Details
Description	Check if user's role grants view access to a specific screen
Actor	System (PermissionGuard)
Input	user.role, screenId

Property	Details
Output	Boolean (allowed/denied)
Tables	screen_permissions, permission_templates
CRUD	R
Business Rules	BR-AUTH-08: Owner role always has full access. BR-AUTH-09: Permission check runs before every page render. BR-AUTH-10: Denied users see "Access Denied" page

Feature 5: Action-Level Permission Check (canAction)

Property	Details
Description	Check if user can perform specific action (create/edit/delete/approve/reject/export/print) on a screen
Actor	System (PermissionGuard)
Input	user.role, screenId, action
Output	Boolean (allowed/denied)
Tables	screen_permissions, permission_templates
CRUD	R
Business Rules	BR-AUTH-11: Buttons disabled if action not permitted. BR-AUTH-12: 8 standard actions: view, create, edit, delete, approve, reject, export, print

Feature 6: Permission Management

Property	Details
Description	Admin UI to assign/revoke permissions per role per screen per action
Actor	Owner, IT Admin
Input	Role, Screen, Action, Granted (boolean)
Output	Updated permission record
Tables	screen_permissions, permission_templates
CRUD	C, R, U, D
Business Rules	BR-AUTH-13: Changes broadcast via Supabase Realtime. BR-AUTH-14: Bulk actions available (Grant All HR / Finance / ALL). BR-AUTH-15: Permission templates can be copied from role to role
Errors	Duplicate permission entry, Invalid screen/action

Feature 7: Permission Templates

Property	Details
Description	Pre-defined permission sets per role for quick assignment
Actor	IT Admin, System (on user creation)
Input	Role name
Output	Auto-applied permission set
Tables	permission_templates
CRUD	C, R, U, D

Property	Details
Business Rules	BR-AUTH-16: New users auto-receive template permissions based on their role

Feature 8: Audit Logging

Property	Details
Description	Record significant system actions for compliance and investigation
Actor	System (automatic)
Input	Action event (user_name, action, details)
Output	Immutable audit record
Tables	audit_log
CRUD	C (insert only — never update/delete)
Business Rules	BR-AUTH-17: Audit logs are append-only. BR-AUTH-18: Logs include IP address

Feature 9: Activity Logging

Property	Details
Description	Detailed activity tracking with old/new value diffs per entity
Actor	System (automatic)
Input	Module, Action, Entity changes
Output	Activity record with JSONB diffs
Tables	activity_log
CRUD	C
Business Rules	BR-AUTH-19: old_values and new_values stored as JSONB for change tracking

Feature 10: Login History

Property	Details
Description	Track all login attempts (success and failure)
Actor	System (automatic)
Input	Login attempt event
Output	Login history record
Tables	login_history
CRUD	C, R
Business Rules	BR-AUTH-20: Both successful and failed attempts recorded. BR-AUTH-21: IP address and user agent captured

Feature 11: QR Code Authentication (Check-in/Out)

Property	Details
Description	One-time QR codes for attendance check-in and check-out

Property	Details
Actor	Employee (scanner), HR (generator)
Input	QR code hash
Output	Attendance record created/updated
Tables	used_qr_codes, attendance
CRUD	C (used_qr_codes), C/U (attendance)
Business Rules	BR-AUTH-22: QR codes are single-use (hash stored after use). BR-AUTH-23: Reused QR codes are rejected
Errors	QR already used, Invalid QR format, Expired QR

Feature 12: Multi-Company Support

Property	Details
Description	Company and branch hierarchy for multi-entity operations
Actor	Owner, IT Admin
Input	Company/Branch data
Output	Scoped data access
Tables	companies, branches, users (company_id, branch_id)
CRUD	C, R, U, D
Business Rules	BR-AUTH-24: Users belong to a company and branch. BR-AUTH-25: Data scoped by company_id

PART 5 & 6 — PROCESSES (Start Actor → End Actor)

Process AUTH-P01: User Login

Property	Details
Process ID	AUTH-P01
Start Actor	Employee / Any User
End Actor	Employee / Any User (receives session)
Trigger	User navigates to login page and submits credentials
Preconditions	User account exists in <code>users</code> table
Main Flow	1. User enters username + password → 2. System queries <code>users</code> WHERE username AND password_hash match → 3. System checks user.status = 'active' → 4. System creates <code>user_sessions</code> record → 5. System inserts <code>login_history</code> (status='success') → 6. System loads permissions from <code>screen_permissions</code> / <code>permission_templates</code> → 7. User redirected to Dashboard
Alternative Flow	3a. If credentials invalid → Insert login_history (status='failed') → Show error. 3b. If status='suspended' → Insert login_history (status='locked') → Show suspended message
Input	username: TEXT, password: TEXT
Output	JWT session token, user profile object, permission array
Tables	users, user_sessions, login_history, screen_permissions, permission_templates

Property	Details
DB Operations	SELECT users, INSERT user_sessions, INSERT login_history, SELECT screen_permissions
Business Rules	BR-AUTH-01, BR-AUTH-02, BR-AUTH-03, BR-AUTH-20, BR-AUTH-21
Permissions	None (public endpoint)
Errors	Invalid credentials, Account suspended, Account inactive, Database connection error
Notifications	None on success. Security alert on multiple failures (future)

Process AUTH-P02: Permission Check

Property	Details
Process ID	AUTH-P02
Start Actor	System (PermissionGuard)
End Actor	System (UI renders or denies)
Trigger	User navigates to any screen or clicks any action button
Main Flow	1. PermissionGuard.canView(screenId) called → 2. Check user.role → 3. Query screen_permissions for (user_id, screen_id, action='view') → 4. If granted → Render page → 5. If denied → Show "Access Denied"
Tables	screen_permissions, permission_templates
DB Operations	SELECT screen_permissions

Process AUTH-P03: QR Code Check-In

Property	Details
Process ID	AUTH-P03
Start Actor	Employee
End Actor	Employee (check-in confirmed)
Trigger	Employee scans QR code at gate
Main Flow	1. Employee scans QR → 2. System extracts qr_hash → 3. System checks used_qr_codes for hash → 4. If NOT used → Insert into used_qr_codes → 5. Create/Update attendance record with check_in timestamp → 6. Show success
Alternative Flow	4a. If QR already used → Reject with "QR Already Used" error
Tables	used_qr_codes, attendance
Errors	QR already used, Invalid QR, Employee not found

Process AUTH-P04: Permission Assignment

Property	Details
Process ID	AUTH-P04
Start Actor	IT Admin / Owner
End Actor	Target User (permissions updated in real-time)

Property	Details
Trigger	Admin opens Permission Management screen and modifies permissions
Main Flow	1. Admin selects user role → 2. Permission grid loads (modules × screens × actions) → 3. Admin toggles permissions → 4. System UPSERTS screen_permissions → 5. Supabase Realtime broadcasts change → 6. Target user's UI updates immediately
Tables	screen_permissions, permission_templates
Notifications	Real-time permission update via Supabase Realtime

Process AUTH-P05: Audit Trail Recording

Property	Details
Process ID	AUTH-P05
Start Actor	Any User (performing an action)
End Actor	System (audit record stored)
Trigger	Any significant action (CRUD operations, approvals, status changes)
Main Flow	1. User performs action → 2. System captures: user_name, action, entity_type, old_values, new_values, IP → 3. System inserts into activity_log → 4. If critical action, also insert into audit_log
Tables	activity_log, audit_log

PART 8 — PROCESS DIAGRAMS

Login Process Flow

```
flowchart TD
    A[👤 User] --> B[Enter Username + Password]
    B --> C[Validate Credentials]
    C -->|Valid| D[Check User Status]
    C -->|Invalid| E[Record Failed Login]
    E --> F[Show Error Message]
    F --> A
    D -->|Active| G[Create Session]
    D -->|Suspended/Inactive| H[Record Locked Login]
    H --> I[Show Suspended Message]
    G --> J[(user_sessions DB)]
    G --> K[(login_history DB)]
    J --> L[Load Permissions]
    L --> M[(screen_permissions DB)]
    M --> N[Redirect to Dashboard]
    N --> O[👤 User - Authenticated]
```

Permission Check Flow

```
flowchart TD
    A[👤 User Navigates to Screen] --> B[PermissionGuard.canView]
    B --> C[Check user.role]
    C -->|Owner| D[✅ Full Access]
    C -->|Other| E[Query screen_permissions]
    E -->|granted=true| F[✅ Render Page]
    E -->|granted=false/missing| G[❌ Access Denied]
    F --> H[Check Action Permissions]
    H --> I[canAction per button]
    I -->|granted| J[Enable Button]
    I -->|denied| K[Disable Button]
```

QR Check-In Flow

```
flowchart TD
    A[👤 Employee] --> B[Scan QR Code]
    B --> C[Extract QR Hash]
    C --> D[Check used_qr_codes]
    D -->|Not Used| E[Insert used_qr_codes]
    D -->|Already Used| F[❌ Reject - QR Reused]
    E --> G[Create/Update Attendance Record]
    G --> H[(attendance DB)]
    H --> I[✅ Check-In Confirmed]
    I --> J[👤 Employee]
```

PART 10 & 11 — RELATIONSHIPS & CARDINALITY

Entity A	Relationship	Entity B	Cardinality	Optionality	FK Location	Business Meaning
users	places	login_history	1:N	Optional (user may never fail login)	login_history.user_id	One user can have zero or many login records
users	has	user_sessions	1:N	Optional	user_sessions.user_id	One user can have zero or many sessions
users	generates	activity_log	1:N	Optional	activity_log.user_id	One user can generate zero or many activity logs
users	generates	audit_log	1:N	Optional	audit_log.user_id	One user can generate zero or many audit logs
users	has	screen_permissions	1:N	Optional	screen_permissions.user_id	One user can have zero or many screen permissions
users	belongs to	companies	N:1	Optional	users.company_id	Many users belong to one company
users	belongs to	branches	N:1	Optional	users.branch_id	Many users belong to one branch
companies	has	branches	1:N	Optional	branches.company_id	One company has zero or many branches
users	uses	used_qr_codes	1:N	Optional	used_qr_codes.used_by	One user can use zero or many QR codes

Crow's Foot Notation

```
USER ||--o{ LOGIN_HISTORY : "generates"
USER ||--o{ USER_SESSIONS : "has"
USER ||--o{ ACTIVITY_LOG : "generates"
USER ||--o{ AUDIT_LOG : "generates"
USER ||--o{ SCREEN_PERMISSIONS : "has"
USER }o--|| COMPANIES : "belongs to"
USER }o--|| BRANCHES : "belongs to"
COMPANIES ||--o{ BRANCHES : "has"
USER ||--o{ USED_QR_CODES : "uses"
```

- `||` = Exactly One (mandatory)
- `o{` = Zero or Many (optional many)
- `}o` = Zero or One (optional one)

PART 12 — OPTIONALITY

Relationship	Mandatory Side	Optional Side	Explanation
users → login_history	login_history MUST have a user	user MAY have zero login records	A login record requires a user, but a user might not have logged in yet
users → user_sessions	user_sessions MUST have a user	user MAY have zero sessions	A session requires a user, but not all users have active sessions
companies → branches	branches MUST belong to a company	company MAY have zero branches	A branch must have a parent company
users → companies	company is optional for user		Multi-company is optional

PART 13 — M:N RELATIONSHIPS

role_permissions (Junction Table)

The relationship between **roles** and **permissions** is M:N: - One role can have many permissions - One permission can be granted to many roles

Junction Table: `role_permissions`

Attribute	Type	Description
id	UUID PK	Primary key
role	TEXT	Role name (FK concept)
permission_id	UUID FK	FK → permissions(id)
granted	BOOLEAN	Whether permission is granted

Why junction is required: A role like "hr manager" needs dozens of permissions, and a permission like "view employees" is shared across multiple roles.

permission_templates (Junction Table)

The relationship between **roles** and **screens+actions** is M:N:

Attribute	Type	Description
id	UUID PK	Primary key
role	TEXT	Role name
screen_id	TEXT	Screen identifier
action	TEXT	Action type
granted	BOOLEAN	Permission granted
UNIQUE		(role, screen_id, action)

PART 14 & 15 — SECTION ERD

```

erDiagram
    USERS ||--o{ LOGIN_HISTORY : "generates"
    USERS ||--o{ USER_SESSIONS : "has"
    USERS ||--o{ ACTIVITY_LOG : "generates"
    USERS ||--o{ AUDIT_LOG : "generates"
    USERS ||--o{ SCREEN_PERMISSIONS : "has"
    USERS ||--o{ USED_QR_CODES : "uses"
    USERS }o--|| COMPANIES : "belongs_to"
    USERS }o--|| BRANCHES : "belongs_to"
    COMPANIES ||--o{ BRANCHES : "has"
    PERMISSIONS ||--o{ ROLE_PERMISSIONS : "granted_via"
    PERMISSION_TEMPLATES }o--|| USERS : "applied_to"

    USERS {
        uuid id PK
        text employee_id UK
        text full_name
        text email UK
        text username UK
        text password_hash
        text role
        text department
        text status
        uuid company_id FK
        uuid branch_id FK
        text preferred_language
        timestampz created_at
    }

    LOGIN_HISTORY {
        uuid id PK
        uuid user_id FK
        text user_name
        text ip_address
        text user_agent
        text login_status
        text failure_reason
        timestampz created_at
    }

    USER_SESSIONS {
        uuid id PK
        uuid user_id FK
        text token_hash
        text refresh_token_hash
        timestampz expires_at
        boolean is_active
        text ip_address
    }

    ACTIVITY_LOG {
        uuid id PK
        uuid user_id FK
        text module
        text action
        text entity_type
        uuid entity_id
        jsonb old_values
  
```

```
        jsonb new_values
        text ip_address
    }

AUDIT_LOG {
    uuid id PK
    text action
    text user_name
    uuid user_id FK
    text details
    text ip
    timestamptz timestamp
}

SCREEN_PERMISSIONS {
    uuid id PK
    uuid user_id FK
    text screen_id
    text action
    boolean granted
    text module
}

PERMISSION_TEMPLATES {
    uuid id PK
    text role
    text screen_id
    text action
    boolean granted
    text module
}

PERMISSIONS {
    uuid id PK
    text module
    text action
    text description_en
    text description_ar
}

ROLE_PERMISSIONS {
    uuid id PK
    text role
    uuid permission_id FK
    boolean granted
}

COMPANIES {
    uuid id PK
    text name
    text name_ar
    text tax_id
    boolean is_active
}

BRANCHES {
    uuid id PK
    uuid company_id FK
    text name
    text name_ar
    boolean is_active
}
```

```
USED_QR_CODES {
  text qr_hash PK
  uuid used_by FK
  timestamptz used_at
  text action_type
}
```

PART 16 — CRUD ANALYSIS

Feature	C	R	U	D	Actor	Tables
User Login	✓	✓			Any User	login_history, user_sessions, users
User Logout			✓		Any User	user_sessions
View Login History		✓			IT Admin, Owner	login_history
View Activity Log		✓			IT Admin, Owner	activity_log
View Audit Log		✓			IT Admin, Owner	audit_log
Manage Permissions	✓	✓	✓	✓	IT Admin, Owner	screen_permissions
Permission Templates	✓	✓	✓	✓	IT Admin	permission_templates
QR Check-In	✓	✓	✓		Employee	used_qr_codes, attendance
QR Generation	✓				HR Manager	(QR generated in memory)
Company Management	✓	✓	✓	✓	Owner	companies
Branch Management	✓	✓	✓	✓	Owner	branches
User Creation	✓				HR Manager, Owner	users, permission_templates

PART 17 — PERMISSIONS MATRIX

Feature	Employee	Hall Manager	HR Manager	IT Admin	Owner
Login/Logout	✓	✓	✓	✓	✓
View Own Profile	✓	✓	✓	✓	✓
View Login History	X	X	✓	✓	✓
View Activity Log	X	X	X	✓	✓
View Audit Log	X	X	✓	✓	✓
Manage Permissions	X	X	X	✓	✓
Permission Templates	X	X	X	✓	✓
QR Check-In/Out	✓	✓	✓	✓	✓
QR Generation	X	X	✓	✓	✓
Company Management	X	X	X	X	✓
System Settings	X	X	X	✓	✓

PART 18 — BUSINESS RULES

Rule ID	Rule	Enforcement
BR-AUTH-01	Username and password must match for login	Application logic
BR-AUTH-02	User status must be 'active' to login	Application logic + DB CHECK
BR-AUTH-03	Failed login attempts must be recorded	Application logic
BR-AUTH-04	Session deactivated on logout	Application logic
BR-AUTH-05	Tokens stored as hashes only	Application logic
BR-AUTH-06	Sessions expire after configured time	Application logic
BR-AUTH-07	Owner role has full access (not bypassing, through permission system)	PermissionGuard
BR-AUTH-08	Permission check runs before every page render	PermissionGuard.canView()
BR-AUTH-09	Denied users see "Access Denied" page	PermissionGuard
BR-AUTH-10	8 standard actions: view, create, edit, delete, approve, reject, export, print	DB seed data
BR-AUTH-11	QR codes are single-use	used_qr_codes PK constraint
BR-AUTH-12	Permission changes broadcast via Realtime	Supabase Realtime
BR-AUTH-13	New users auto-receive role template permissions	Application logic on user creation
BR-AUTH-14	Audit logs are append-only (insert only)	Application logic
BR-AUTH-15	Multi-company data scoped by company_id	Application logic

PART 21 — AUDIT AND HISTORY

Log Type	Table	Purpose	Retention
Login History	login_history	Track all login attempts (success/failure)	Indefinite
Activity Log	activity_log	Track entity-level changes with old/new JSONB diffs	Indefinite
Audit Log	audit_log	Track significant system actions	Indefinite
Session History	user_sessions	Track active/expired sessions	Indefinite
QR Usage Log	used_qr_codes	Track QR code usage to prevent reuse	Indefinite

PART 22 — NOTIFICATIONS

Event	Sender	Receiver	Notification Type
Permission Changed	System	Affected User	Real-time (Supabase Realtime)
Login from New IP	System	User / IT Admin	Alert (future)
Multiple Failed Logins	System	IT Admin	Security Alert (future)
New User Created	System	HR Manager	Notification
Account Suspended	HR Manager	IT Admin	Notification

PART 23 — ERROR AND EXCEPTION FLOWS

Login Failure Flow

```
flowchart TD
    A[👤 User] --> B[Submit Credentials]
    B --> C{Validate}
    C -->|Invalid Password| D[login_history: status=failed]
    C -->|User Not Found| E[login_history: status=failed]
    C -->|Account Suspended| F[login_history: status=locked]
    D --> G[Show: Invalid Credentials]
    E --> G
    F --> H[Show: Account Suspended]
    G --> A
    H --> I[Contact HR]
```

QR Reuse Error Flow

```
flowchart TD
    A[👤 Employee] --> B[Scan QR]
    B --> C{Check used_qr_codes}
    C -->|Hash Exists| D[❌ QR Already Used]
    D --> E[Show Error Message]
    E --> F[Employee Must Get New QR]
    C -->|Hash Not Found| G[✅ Process Check-In]
```

Error	Impact	Database Effect	Recovery
Invalid credentials	Login blocked	login_history record created	User retries or resets password
Account suspended	Login blocked	login_history record created	HR must reactivate account
QR already used	Check-in rejected	No attendance record created	Generate new QR code
Session expired	Access denied	Session marked inactive	User must re-login
Permission denied	Page/Action blocked	No data change	Admin must grant permission
Database connection error	System unavailable	No records	Retry or contact IT

PART 4 — SECTION 2: HR & People Management 👤

4.1 Section Purpose

The HR & People Management module is the **largest and most complex module** in the system. It manages the complete employee lifecycle from recruitment through offboarding, including daily operations like attendance tracking, leave management, and performance evaluation.

- **Why it exists:** Centralizes all HR operations for a polymer manufacturing factory with 16 departments and hundreds of employees across multiple shifts

- **What problem it solves:** Replaces paper-based attendance sheets, manual leave tracking, and disconnected HR processes with an integrated digital system
- **Which users interact:** HR Manager, Hall Managers, Employees (self-service), Owner (final approvals), Nurse (medical integration)
- **What data it manages:** 20+ tables covering employee records, attendance, leaves, absence tracking, overtime, missions, complaints, disciplinary actions, offboarding, performance reviews, training, asset assignments, warnings, Friday work requests, and recruitment/ATS

4.2 Actors

Actor	Can Do	Cannot Do
Employee	View own data, Submit leave/mission/expense/complaint requests, QR check-in/out, View own salary	View other employees' data, Approve requests, Edit attendance
Hall Manager	Approve/Reject team leaves (1st level), View team attendance, Submit Friday work requests	Approve own leaves, Process payroll, Access other departments
HR Manager	Full CRUD on employees, Process attendance, Approve leaves (2nd level), Manage recruitment, Issue warnings, Process offboarding, Manage training	Final approve leaves (Owner), Release payroll funds
Owner	Final leave approval (3rd level), Approve loans, View all HR data	N/A (full access)
Nurse	View employee data for clinic visits, Issue rest permits	Edit employee records, Approve leaves

4.3 Features (Detailed)

Feature Group A: Employee Management

Feature	Description	Actor	Input	Output	Tables	CRUD	Business Rules
Create Employee	Add new employee record	HR Manager	Personal data, role, department, salary	Employee record + auto-permissions	users, permission_templates	C	BR-HR-01: employee_id must be unique. BR-HR-02: Email and username must be unique
View Employee	View employee details	HR Manager, Hall Manager (team only)	employee_id	Employee profile	users	R	BR-HR-03: Employees see only own data
Update Employee	Edit employee information	HR Manager	Updated fields	Updated record + activity_log	users, activity_log	U	BR-HR-04: Salary changes logged
Deactivate Employee	Set employee status to inactive	HR Manager	employee_id, reason	Status changed	users	U	BR-HR-05: Triggers offboarding
View Org Directory	Browse organization structure	All authenticated	-	Employee cards grouped by department	users	R	

Feature Group B: Attendance Management

Feature	Description	Actor	Input	Output	Tables	CRUD
QR Check-In	Employee scans QR at gate entry	Employee	QR hash	Attendance record with check_in time	attendance, used_qr_codes	C
QR Check-Out	Employee scans QR at gate exit	Employee	QR hash	Attendance updated with check_out, working_hours calculated	attendance, used_qr_codes	U
Manual Attendance	HR enters attendance manually	HR Manager	employee, date, check_in, check_out	Attendance record	attendance	C/U
View Attendance	View attendance records	HR (all), Employee (own)	Date range, department filter	Attendance list with stats	attendance	R
Attendance Report	Export attendance summary	HR Manager	Date range	PDF/Excel report	attendance	R
Late Detection	System calculates delay_minutes	System	check_in time vs shift start	delay_minutes field, possible late_deduction	attendance, late_deductions	U/C
Generate QR Code	HR generates single-use QR	HR Manager	-	QR code image	(in-memory generation)	-

Attendance Business Rules: - BR-HR-06: $\text{delay_minutes} = \text{check_in} - \text{shift_start_time}$ (if positive) - BR-HR-07: $\text{working_hours} = \text{check_out} - \text{check_in}$ (in hours) - BR-HR-08: Status transitions: absent → checked_in → present - BR-HR-09: QR codes are single-use (enforced by used_qr_codes PK)

Feature Group C: Leave Management

Feature	Description	Actor	Input	Output	Tables	CRUD
Submit Leave Request	Employee requests time off	Employee	type, start_date, end_date, reason	Leave request (status=pending_manager)	leave_requests	C
Manager Approve Leave	Hall Manager reviews request	Hall Manager	request_id, approve/reject	Status → pending_hr OR rejected	leave_requests	U
HR Approve Leave	HR reviews manager-approved request	HR Manager	request_id, approve/reject	Status → pending_owner OR rejected	leave_requests	U
Owner Approve Leave	Final approval	Owner	request_id, approve/reject	Status → approved OR rejected	leave_requests	U
View My Leaves	Employee views own leave history	Employee	-	Leave list with balances	leave_requests	R
Manage All Leaves	HR views/manages all leave requests	HR Manager	Filters	Leave dashboard	leave_requests	R/U

Leave Approval Workflow:

Employee → pending_manager → Hall Manager → pending_hr → HR Manager → pending_owner → Owner → approved/rejected

Leave Business Rules: - BR-HR-10: Leave types: Annual, Sick, Emergency, Unpaid, Maternity/Paternity - BR-HR-11: Annual leave deducted from annual_leave_balance (default 24 days/year) - BR-HR-12: Emergency leaves may bypass manager approval - BR-HR-13: Rejected requests must include rejection_reason - BR-HR-14: Approved annual leaves reduce annual_leave_balance by days count

Feature Group D: Absence Management (Advanced)

Feature	Description	Actor	Tables	CRUD
Submit Absence Request	Employee submits pre-planned absence	Employee	hr_absence_requests	C
Manager Review Absence	Manager approves/rejects absence	Hall Manager	hr_absence_requests	U
HR Review Absence	HR final decision	HR Manager	hr_absence_requests	U
Record Unauthorized Absence	System/HR records unexcused absence	HR Manager	hr_absence_records	C
Override Absence Penalty	HR changes default 2-day penalty	HR Manager	hr_absence_records, hr_absence_audit	U
Auto-Warning Trigger	System triggers warning for consecutive absences	System	hr_violations_warnings	C
Manage Absence Settings	Configure penalty defaults	HR Manager	hr_absence_settings	U

Absence Business Rules: - BR-HR-15: Default penalty for unauthorized absence = 2 days salary deduction - BR-HR-16: HR can override to: deduct_2, deduct_1, casual_leave, unpaid_leave, cancelled - BR-HR-17: 3 consecutive unauthorized absences → first_warning - BR-HR-18: Continued absence after warning → final_warning - BR-HR-19: Continued absence after final warning → termination - BR-HR-20: All absence changes logged in hr_absence_audit

Feature Group E: Overtime & Missions

Feature	Description	Actor	Tables	CRUD
Submit Overtime	Record overtime hours	Employee / HR	overtime	C
Approve Overtime	Manager approves overtime	Hall Manager	overtime	U
Submit Mission/Errand	Employee requests to leave for errand	Employee	missions	C
Approve Mission	Manager approves mission	Hall Manager	missions	U
QR Mission Out	Employee scans QR leaving for mission	Employee	missions, used_qr_codes	U
QR Mission Return	Employee scans QR returning from mission	Employee	missions, used_qr_codes	U
Friday Work Request	Department requests employees work on Friday	Hall Manager	friday_work_requests	C
Approve Friday Work	HR approves Friday work	HR Manager	friday_work_requests	U

Business Rules: - BR-HR-21: Overtime rate default = 1.5x base hourly rate - BR-HR-22: Overtime must be approved before payroll calculation - BR-HR-23: Missions track actual_time_out and actual_time_in via QR - BR-HR-24: Friday work requests include employee UUID array

Feature Group F: Complaints & Disciplinary

Feature	Description	Actor	Tables	CRUD
Submit Complaint	Employee files complaint (can be anonymous)	Employee	complaints	C
Resolve Complaint	HR resolves complaint	HR Manager	complaints	U
Issue Disciplinary Action	HR issues disciplinary action	HR Manager	disciplinary_actions	C
Issue Warning	HR issues formal warning	HR Manager	employee_warnings	C
View Warnings	View employee warnings history	HR Manager / Employee	employee_warnings	R

Business Rules: - BR-HR-25: Anonymous complaints hide employee identity (is_anonymous=true) - BR-HR-26: Warning types escalate: verbal → written → final → suspension → termination - BR-HR-27: Warnings can be appealed (status →

appealed) - BR-HR-28: Warnings can include deduction_amount and deduction_days

Feature Group G: Performance & Training

Feature	Description	Actor	Tables	CRUD
Create Performance Review	Manager creates review for employee	Hall Manager / HR	performance_reviews	C
Submit Review	Submit completed review	Reviewer	performance_reviews	U
Acknowledge Review	Employee acknowledges review	Employee	performance_reviews	U
Create Training Course	Plan training course	HR Manager	training_courses	C
Enroll Employee	Enroll employee in course	HR Manager	training_enrollments	C
Record Completion	Mark course as completed	HR Manager	training_enrollments	U
Assign Asset	Assign company asset to employee	HR Manager	asset_assignments	C
Return Asset	Record asset return	HR Manager	asset_assignments	U

Business Rules: - BR-HR-29: Performance scores are 1-5 for each dimension (attendance, quality, teamwork, initiative, communication) - BR-HR-30: Review workflow: draft → submitted → reviewed → acknowledged - BR-HR-31: Training types: internal, external, online - BR-HR-32: Asset types: laptop, phone, tablet, vehicle, key, uniform, tool, other

Feature Group H: Offboarding

Feature	Description	Actor	Tables	CRUD
Initiate Offboarding	Start employee separation process	HR Manager	offboarding	C
IT Clearance	IT confirms equipment returned	IT Admin	offboarding (it_cleared)	U
HR Clearance	HR confirms HR obligations cleared	HR Manager	offboarding (hr_cleared)	U
Finance Clearance	Finance confirms financial obligations cleared	Accountant	offboarding (finance_cleared)	U
Complete Offboarding	All clearances done, complete process	HR Manager	offboarding, users (status→inactive)	U

Business Rules: - BR-HR-33: All three clearances (IT, HR, Finance) required before completion - BR-HR-34: Offboarding deactivates user account (status → inactive)

Feature Group I: Recruitment / ATS

Feature	Description	Actor	Tables	CRUD
Post Job	Create job posting	HR Manager	(handled via ats_applications context)	-
Submit Application	Candidate applies with CV	External / HR	ats_applications	C
AI CV Scoring	AI analyzes CV and scores candidate	AI Engine	ats_applications (ai_score, ai_verdict)	U
Review Application	HR reviews AI-scored application	HR Manager	ats_applications	U
Schedule Interview	Move candidate to interview stage	HR Manager	ats_applications (status→interview)	U
Hire Candidate	Convert to employee	HR Manager	ats_applications (status→hired), users	U/C

Business Rules: - BR-HR-35: AI verdict: accepted, rejected, or review - BR-HR-36: Application workflow: pending → screening → interview → offered → hired / rejected - BR-HR-37: AI matches skills from CV against job requirements

PART 5 & 6 — PROCESSES

Process HR-P01: Leave Request Approval Cycle

Property	Details
Process ID	HR-P01
Start Actor	Employee
End Actor	Employee (receives approval/rejection notification)
Trigger	Employee submits leave request
Main Flow	1. Employee submits leave request → 2. Status = pending_manager → 3. Hall Manager reviews → 4. If approved: status = pending_hr → 5. HR Manager reviews → 6. If approved: status = pending_owner → 7. Owner reviews → 8. If approved: status = approved, annual_leave_balance decreased → 9. Notification sent to Employee
Alternative Flows	3a. Manager rejects → status = rejected, notification to Employee. 5a. HR rejects → same. 7a. Owner rejects → same
Tables	leave_requests, users (annual_leave_balance), notifications
Business Rules	BR-HR-10 through BR-HR-14
Errors	Insufficient leave balance, Invalid date range, Leave conflict
Notifications	Each approval/rejection generates notification to employee

Process HR-P02: Attendance Check-In/Out

Property	Details
Process ID	HR-P02
Start Actor	Employee
End Actor	System (attendance recorded)
Trigger	Employee scans QR code at factory gate
Main Flow	1. Employee scans QR → 2. System validates QR hash → 3. If check-in: Create attendance record with check_in, calculate delay_minutes → 4. If check-out: Update attendance with check_out, calculate working_hours → 5. If late: Create late_deductions record
Tables	used_qr_codes, attendance, late_deductions

Process HR-P03: Unauthorized Absence Handling

Property	Details
Process ID	HR-P03
Start Actor	System (detects absence)
End Actor	Employee (receives penalty/warning)
Trigger	Employee does not check in by shift end
Main Flow	1. System detects no attendance for employee on workday → 2. HR records unauthorized absence → 3. Default 2-day salary penalty applied → 4. System checks consecutive absence count → 5. If ≥ 3 consecutive: Generate first_warning → 6. If continued: final_warning → 7. If continued: termination recommendation
Tables	hr_absence_records, hr_violations_warnings, hr_absence_audit

Property	Details
Business Rules	BR-HR-15 through BR-HR-20

Process HR-P04: Offboarding Clearance

Property	Details
Process ID	HR-P04
Start Actor	HR Manager
End Actor	HR Manager (process completed)
Trigger	Employee resignation/termination
Main Flow	1. HR initiates offboarding → 2. IT receives clearance request → 3. IT confirms equipment returned (it_cleared=true) → 4. Finance confirms no pending dues (finance_cleared=true) → 5. HR confirms HR obligations clear (hr_cleared=true) → 6. All cleared → status=completed → 7. User account deactivated
Tables	offboarding, users

Process HR-P05: ATS Recruitment with AI

Property	Details
Process ID	HR-P05
Start Actor	Candidate (submits CV) or HR Manager (enters CV)
End Actor	HR Manager (hires or rejects)
Main Flow	1. CV submitted → 2. AI Engine analyzes cv_text → 3. AI calculates ai_score, sets ai_verdict → 4. HR reviews AI analysis → 5. If promising: status → screening → interview → offered → hired → 6. Create user account
Tables	ats_applications, users

PART 8 — PROCESS DIAGRAMS

Leave Approval Workflow

```

flowchart TD
    A[👤 Employee] --> B[Submit Leave Request]
    B --> C[Status: pending_manager]
    C --> D[Hall Manager Review]
    D -->|Approve| E[Status: pending_hr]
    D -->|Reject| F[Status: rejected]
    E --> G[HR Manager Review]
    G -->|Approve| H[Status: pending_owner]
    G -->|Reject| F
    H --> I[Owner Review]
    I -->|Approve| J[Status: approved]
    I -->|Reject| F
    J --> K[Deduct from annual_leave_balance]
    K --> L[📧 Notify Employee]
    F --> L
    L --> M[👤 Employee]
  
```

Attendance Flow

```

flowchart TD
    A[👤 Employee] --> B[Scan QR at Gate]
    B --> C{First Scan Today?}
    C -->|Yes - Check In| D[Create Attendance Record]
    C -->|No - Check Out| E[Update Attendance Record]
    D --> F{Late?}
    F -->|Yes| G[Calculate delay_minutes]
    G --> H[Create late_deductions]
    F -->|No| I[Status: present]
    E --> J[Calculate working_hours]
    J --> K[Status: present]
    H --> I
  
```

Absence Penalty Escalation

```

flowchart TD
    A[System Detects Absence] --> B[HR Records Absence]
    B --> C[Apply 2-Day Penalty Default]
    C --> D{HR Override?}
    D -->|Yes| E[Apply HR Decision]
    D -->|No| F[Keep Default Penalty]
    E --> G{Check Consecutive Days}
    F --> G
    G -->|< 3 days| H[Record Only]
    G -->|>= 3 days| I[Issue First Warning]
    I --> J{Continued Absence?}
    J -->|Yes| K[Issue Final Warning]
    K --> L{Still Absent?}
    L -->|Yes| M[Recommend Termination]
    L -->|No| N[Employee Returns]
  
```

Offboarding Flow

flowchart TD

```

A[👤 HR Manager] --> B[Initiate Offboarding]
B --> C[IT Clearance Request]
B --> D[Finance Clearance Request]
B --> E[HR Clearance Request]
C --> F{IT Cleared?}
D --> G{Finance Cleared?}
E --> H{HR Cleared?}
F -->|✓| I[it_cleared = true]
G -->|✓| J[finance_cleared = true]
H -->|✓| K[hr_cleared = true]
I --> L{All Three Cleared?}
J --> L
K --> L
L -->|Yes| M[Status: completed]
M --> N[Deactivate User Account]
N --> O[👤 HR Manager - Confirmed]

```

ATS Recruitment Flow

flowchart TD

```

A[📄 Candidate CV] --> B[Submit Application]
B --> C[🤖 AI Engine Analyzes CV]
C --> D[AI Score + Verdict]
D --> E{AI Verdict}
E -->|Accepted| F[Status: screening]
E -->|Review| G[HR Manual Review]
E -->|Rejected| H[Status: rejected]
G -->|Pass| F
G -->|Fail| H
F --> I[Schedule Interview]
I --> J{Interview Result}
J -->|Pass| K[Status: offered]
J -->|Fail| H
K --> L{Accept Offer?}
L -->|Yes| M[Status: hired]
L -->|No| H
M --> N[Create Employee Record]
N --> O[👤 HR Manager]

```

PART 14 & 15 — SECTION ERD

```

erDiagram
    USERS ||--o{ ATTENDANCE : "tracks"
    USERS ||--o{ LEAVE_REQUESTS : "submits"
    USERS ||--o{ OVERTIME : "records"
    USERS ||--o{ MISSIONS : "requests"
    USERS ||--o{ COMPLAINTS : "files"
    USERS ||--o{ DISCIPLINARY_ACTIONS : "receives"
    USERS ||--o{ OFFBOARDING : "undergoes"
    USERS ||--o{ EMPLOYEE_WARNINGS : "receives"
    USERS ||--o{ PERFORMANCE_REVIEWS : "reviewed_in"
    USERS ||--o{ TRAINING_ENROLLMENTS : "enrolled_in"
    USERS ||--o{ ASSET_ASSIGNMENTS : "assigned"
    USERS ||--o{ HR_ABSENCE_REQUESTS : "submits"
    USERS ||--o{ HR_ABSENCE_RECORDS : "has"
    USERS ||--o{ HR_VIOLATIONS_WARNINGS : "receives"
    USERS ||--o{ FRIDAY_WORK_REQUESTS : "requests"
    TRAINING_COURSES ||--o{ TRAINING_ENROLLMENTS : "has"
    USERS ||--o{ ATS_APPLICATIONS : "reviews"

    USERS {
        uuid id PK
        text employee_id UK
        text full_name
        text email UK
        text role
        text department
        text position
        numeric base_salary
        text shift
        text status
        numeric annual_leave_balance
        date hire_date
        boolean insurance_active
    }

    ATTENDANCE {
        uuid id PK
        uuid employee_id FK
        text employee_name
        text department
        date date
        timestamptz check_in
        timestamptz check_out
        text shift
        integer delay_minutes
        numeric working_hours
        text status
        text qr_check_in
        text qr_check_out
    }

    LEAVE_REQUESTS {
        uuid id PK
        uuid employee_id FK
        text type
        date start_date
        date end_date
        numeric days
  
```

```
    text reason
    text status
    text rejection_reason
    text approved_by
}

OVERTIME {
    uuid id PK
    uuid employee_id FK
    date date
    numeric hours
    text reason
    text status
    numeric rate
    text approved_by
}

MISSIONS {
    uuid id PK
    uuid employee_id FK
    date mission_date
    text reason
    time time_out
    time time_in
    text status
    text qr_out
    text qr_in
    numeric overtime_hours
}

COMPLAINTS {
    uuid id PK
    uuid employee_id FK
    text subject
    text description
    boolean is_anonymous
    text status
}

DISCIPLINARY_ACTIONS {
    uuid id PK
    uuid employee_id FK
    text type
    text reason
    uuid issued_by FK
}

OFFBOARDING {
    uuid id PK
    uuid employee_id FK
    date separation_date
    text reason
    boolean it_cleared
    boolean hr_cleared
    boolean finance_cleared
    text status
}

LATE_DEDUCTIONS {
    uuid id PK
    text employee_name
    date date
}
```

```
integer delay_minutes
text deduction_type
numeric amount
}

HR_ABSENCE_REQUESTS {
  uuid id PK
  uuid employee_id
  text type
  date start_date
  date end_date
  boolean emergency_flag
  text status
  text manager_notes
  text hr_notes
}

HR_ABSENCE_RECORDS {
  uuid id PK
  uuid employee_id
  date date
  text type
  numeric penalty_days
  text hr_decision
  text status
}

HR_VIOLATIONS_WARNINGS {
  uuid id PK
  uuid employee_id
  text warning_level
  text reason
  integer consecutive_days
  text status
}

HR_ABSENCE_AUDIT {
  uuid id PK
  uuid record_id
  text table_name
  text action
  text old_value
  text new_value
  text changed_by
}

PERFORMANCE_REVIEWS {
  uuid id PK
  uuid employee_id FK
  uuid reviewer_id FK
  text review_period
  integer overall_rating
  integer attendance_score
  integer quality_score
  text status
}

TRAINING_COURSES {
  uuid id PK
  text title
  text trainer
  text course_type
}
```

```
text department
date start_date
date end_date
text status
}

TRAINING_ENROLLMENTS {
  uuid id PK
  uuid course_id FK
  uuid employee_id FK
  text status
  numeric score
  text certificate_url
}

ASSET_ASSIGNMENTS {
  uuid id PK
  text asset_name
  text asset_type
  text serial_number
  uuid assigned_to FK
  uuid assigned_by FK
  date assigned_date
  date return_date
  text status
}

EMPLOYEE_WARNINGS {
  uuid id PK
  uuid employee_id FK
  text warning_type
  text reason
  date incident_date
  uuid issued_by FK
  numeric deduction_amount
  text status
}

FRIDAY_WORK_REQUESTS {
  uuid id PK
  text department
  uuid requested_by FK
  uuid_arr employees
  date request_date
  text status
}

ATS_APPLICATIONS {
  uuid id PK
  text job_title
  text department
  text candidate_name
  text candidate_email
  text cv_text
  numeric ai_score
  text ai_verdict
  text status
}
```

PART 16 — CRUD ANALYSIS

Feature	C	R	U	D	Actor	Tables
Create Employee	✓				HR Manager	users
View Employees		✓			HR Manager	users
Update Employee			✓		HR Manager	users
Delete Employee				✓	Owner	users
QR Check-In	✓	✓			Employee	attendance, used_qr_codes
QR Check-Out		✓	✓		Employee	attendance, used_qr_codes
View My Attendance		✓			Employee	attendance
View All Attendance		✓			HR Manager	attendance
Submit Leave	✓				Employee	leave_requests
Approve Leave (Manager)		✓	✓		Hall Manager	leave_requests
Approve Leave (HR)		✓	✓		HR Manager	leave_requests
Approve Leave (Owner)		✓	✓		Owner	leave_requests
Submit Overtime	✓				Employee	overtime
Approve Overtime		✓	✓		Hall Manager	overtime
Submit Mission	✓				Employee	missions
Submit Complaint	✓				Employee	complaints
Resolve Complaint		✓	✓		HR Manager	complaints
Issue Warning	✓				HR Manager	employee_warnings
Create Performance Review	✓				Hall Manager/HR	performance_reviews
Create Training	✓				HR Manager	training_courses
Enroll in Training	✓				HR Manager	training_enrollments
Assign Asset	✓				HR Manager	asset_assignments
Return Asset			✓		HR Manager	asset_assignments
Initiate Offboarding	✓				HR Manager	offboarding
Complete Offboarding			✓		HR/IT/Finance	offboarding
Submit ATS Application	✓				HR/Candidate	ats_applications
Review ATS Application		✓	✓		HR Manager	ats_applications
Record Absence	✓				HR Manager	hr_absence_records
Override Absence Penalty			✓		HR Manager	hr_absence_records
Friday Work Request	✓				Hall Manager	friday_work_requests

PART 17 — PERMISSIONS MATRIX

Feature	Employee	Hall Mgr	HR Mgr	Owner
View Own Profile	✓	✓	✓	✓
View All Employees	X	Team	✓	✓
Create Employee	X	X	✓	✓
Edit Employee	X	X	✓	✓
Delete Employee	X	X	X	✓
View Own Attendance	✓	✓	✓	✓
View All Attendance	X	Team	✓	✓
Submit Leave	✓	✓	✓	✓
Approve Leave (L1)	X	✓	X	X
Approve Leave (L2)	X	X	✓	X
Approve Leave (L3)	X	X	X	✓
Submit Overtime	✓	✓	✓	✓
Approve Overtime	X	✓	✓	✓
Submit Complaint	✓	✓	✓	✓
Resolve Complaint	X	X	✓	✓
Issue Warning	X	X	✓	✓
Manage Recruitment	X	X	✓	✓
Process Offboarding	X	X	✓	✓
Manage Training	X	X	✓	✓
Performance Reviews	X	Create	✓	✓
Absence Management	X	X	✓	✓
Friday Work Requests	X	✓	✓	✓
Export Reports	X	X	✓	✓

PART 18 — BUSINESS RULES SUMMARY

ID	Rule	Impact
BR-HR-01	employee_id must be unique	Prevents duplicate employee codes
BR-HR-02	Email and username must be unique	Prevents duplicate accounts
BR-HR-03	Employees can only view their own data	Privacy and security
BR-HR-06	Delay = check_in - shift_start (if positive)	Auto-calculate lateness
BR-HR-07	working_hours = check_out - check_in	Auto-calculate work duration
BR-HR-10	Leave types: Annual/Sick/Emergency/Unpaid/Maternity	Constrained by DB CHECK
BR-HR-11	Annual leave deducted from balance (default 24 days)	Balance tracking

ID	Rule	Impact
BR-HR-13	Rejected requests must include rejection_reason	Accountability
BR-HR-15	Default unauthorized absence penalty = 2 days salary	Standard penalty
BR-HR-17	3+ consecutive absences → first_warning	Escalation policy
BR-HR-25	Anonymous complaints hide employee identity	Employee protection
BR-HR-26	Warning escalation: verbal→written→final→suspension→termination	Progressive discipline
BR-HR-29	Performance scores 1-5 per dimension	Standardized evaluation
BR-HR-33	All 3 clearances required for offboarding completion	Complete exit process
BR-HR-35	AI CV scoring: accepted/rejected/review	Automated screening

PART 22 — NOTIFICATIONS

Event	Sender	Receiver	Type
Leave Request Submitted	Employee	Hall Manager	Notification
Leave Approved by Manager	Hall Manager	HR Manager	Notification
Leave Approved by HR	HR Manager	Owner	Notification
Leave Final Decision	Owner / HR / Manager	Employee	Notification
Overtime Submitted	Employee	Hall Manager	Notification
Warning Issued	HR Manager	Employee	Notification
Absence Penalty Applied	System	Employee	Notification
Offboarding Initiated	HR Manager	IT/Finance	Notification
Training Enrollment	HR Manager	Employee	Notification
Performance Review Created	Reviewer	Employee	Notification
ATS New Application	System/External	HR Manager	Notification
Friday Work Approved	HR Manager	Hall Manager	Notification

PART 23 — ERROR FLOWS

Error Scenario	Impact	Database Effect	Recovery
Leave request with insufficient balance	Request blocked	No record created	Employee must choose Unpaid leave
Duplicate attendance for same date	Insert fails	No change	Check existing record first
QR code already used	Check-in rejected	No attendance update	Generate new QR
Leave date conflict (overlapping)	Request should be rejected	No record created	Employee adjusts dates
Offboarding without all clearances	Cannot complete	Status stays pending	Complete remaining clearances
Invalid leave type	Insert fails	CHECK constraint violation	Use valid type from enum
Absence penalty override without reason	Audit incomplete	Override recorded without context	HR must provide reason

PART 4 — SECTION 3: Payroll & Compensation

4.1 Section Purpose

Automates monthly salary calculation integrating data from Attendance, Overtime, Leaves, Loans, and Late Deductions modules. Manages employee loans, medical reimbursements, and expense claims.

- **Why:** Manual payroll calculation for hundreds of factory workers across shifts is error-prone and time-consuming
- **Problem Solved:** Automated net salary = base_salary + overtime_pay + bonuses + performance_bonus - penalties - late_deductions - absence_deductions - loan_installment
- **Users:** HR Manager (calculate/process), Owner (approve funding), Employee (view own), Accountant (payment processing)

4.2 Actors

Actor	Role in This Section
HR Manager	Calculates payroll, manages adjustments, processes loans
Owner	Approves payroll funding, approves loans, approves medical disbursements
Employee	Views own salary, submits loan/medical/expense requests
Accountant	Processes final payments, records in GL

4.3 Features

Feature	Actor	Input	Output	Tables	CRUD	Business Rules
Calculate Monthly Payroll	HR Manager	Month (YYYY-MM)	Payroll records for all employees	payroll, attendance, overtime, loans, late_deductions	C	BR-PAY-01: Net = Base + OT + Bonuses - Penalties - Deductions
View Payroll Dashboard	HR Manager	Month filter	Summary with totals	payroll	R	
Release Funds	Owner	Month	Status → funds_released	payroll	U	BR-PAY-02: Owner must approve before payment
Mark as Paid	HR Manager	Month	Status → paid, paid_date set	payroll	U	BR-PAY-03: Funds must be released first
View My Salary	Employee	-	Own payroll history	payroll	R	BR-PAY-04: Employee sees only own records
Salary Adjustments	HR Manager	Employee, amount, type	Adjustment record	salary_adjustments	C/R/U	
Submit Loan Request	Employee	Amount, installments, reason	Loan request (status=pending)	loans	C	BR-PAY-05: Monthly_deduction = amount / installments

Feature	Actor	Input	Output	Tables	CRUD	Business Rules
Approve Loan	Owner	loan_id	Status → active	loans	U	BR-PAY-06: Owner approval required
Confirm Loan Receipt	Employee	loan_id	employee_receipt_status → confirmed	loans	U	BR-PAY-07: Employee confirms money received
HR Final Loan Approval	HR Manager	loan_id	hr_final_approval = true	loans	U	
Defer Loan Month	HR Manager	loan_id, month	Month added to deferred_months[]	loans	U	BR-PAY-08: Deferred months skip deduction
Complete Loan	System	remaining = 0	Status → completed	loans	U	BR-PAY-09: Auto-complete when fully paid
Submit Medical Request	Employee	Description, document, amount	Request (status=pending)	medical_requests	C	
Approve Medical	Owner	request_id	Status → approved_by_owner	medical_requests	U	BR-PAY-10: Owner approval required
Disburse Medical	HR Manager	request_id	Status → disbursed	medical_requests	U	
Submit Expense Claim	Employee	Type, amount, date, description	Expense (status=pending)	expenses	C	
Approve Expense	HR Manager	expense_id	Status → approved	expenses	U	
Late Deduction Calc	System	delay_minutes	Deduction record	late_deductions	C	BR-PAY-11: Deduction based on delay_minutes threshold

Processes

Process PAY-P01: Monthly Payroll Calculation

flowchart TD

```

A[👤 HR Manager] --> B[Select Month]
B --> C[Fetch All Active Employees]
C --> D[For Each Employee]
D --> E[Get base_salary from users]
E --> F[Calculate overtime_pay from approved overtime]
F --> G[Get bonuses + performance_bonus]
G --> H[Calculate late_deductions from attendance delays]
H --> I[Calculate absence_deductions from absences]
I --> J[Get active loan monthly_deduction]
J --> K{Deferred Month?}
K -->|Yes| L[Skip Loan Deduction]
K -->|No| M[Include Loan Deduction]
L --> N[Net = Base + OT + Bonuses - Penalties - Late - Absence - Loan]
M --> N
N --> O[Insert Payroll Record]
O --> P[(payroll DB)]
P --> Q[Status: processing]
Q --> R[👤 HR Manager - Review]

```

Start Actor: HR Manager | **End Actor:** HR Manager (reviews generated payroll)

Process PAY-P02: Loan Lifecycle

flowchart TD

```

A[👤 Employee] --> B[Submit Loan Request]
B --> C[Status: pending]
C --> D[Owner Approval]
D -->|Approve| E[Status: active]
D -->|Reject| F[Status: rejected]
E --> G[👤 Employee Confirms Receipt]
G --> H[HR Final Approval]
H --> I[Monthly Deduction Begins]
I --> J[Each Month]
J --> K{Deferred?}
K -->|No| L[Deduct from Payroll]
K -->|Yes| M[Skip This Month]
L --> N[remaining_amount -= monthly_deduction]
N --> O{remaining = 0?}
O -->|Yes| P[Status: completed]
O -->|No| J
F --> Q[👤 Employee Notified]
P --> Q
  
```

Start Actor: Employee | **End Actor:** Employee (loan completed or rejected)

Process PAY-P03: Medical Request

flowchart TD

```

A[👤 Employee] --> B[Submit Medical Request + Document]
B --> C[Status: pending]
C --> D[Owner Review]
D -->|Approve| E[Status: approved_by_owner]
D -->|Reject| F[Status: rejected]
E --> G[HR Processes Disbursement]
G --> H[Status: disbursed]
H --> I[👤 Employee Receives Payment]
  
```

Start Actor: Employee | **End Actor:** Employee (receives disbursement)

Section ERD

```
erDiagram
    USERS ||--o{ PAYROLL : "has"
    USERS ||--o{ LOANS : "requests"
    USERS ||--o{ MEDICAL_REQUESTS : "submits"
    USERS ||--o{ EXPENSES : "claims"
    USERS ||--o{ LATE_DEDUCTIONS : "incurs"

    PAYROLL {
        uuid id PK
        uuid employee_id FK
        text month
        numeric base_salary
        numeric overtime_pay
        numeric bonuses
        numeric performance_bonus
        numeric penalties
        numeric late_deductions
        numeric absence_deductions
        numeric net_salary
        text status
        date paid_date
    }

    LOANS {
        uuid id PK
        uuid employee_id FK
        numeric amount
        integer installments
        numeric monthly_deduction
        numeric remaining_amount
        text status
        text reason
        text_arr deferred_months
        text approved_by
        text employee_receipt_status
        boolean hr_final_approval
    }

    MEDICAL_REQUESTS {
        uuid id PK
        uuid employee_id FK
        text description
        text document_url
        numeric amount
        text status
        text owner_comments
        text hr_comments
        text rejection_reason
    }

    EXPENSES {
        uuid id PK
        uuid employee_id FK
        text type
        numeric amount
        date date
        text description
        text status
```

```
}

LATE_DEDUCTIONS {
  uuid id PK
  text employee_name
  date date
  integer delay_minutes
  text deduction_type
  numeric amount
}
```

CRUD & Permissions

Feature	Employee	HR Mgr	Owner	Accountant
View Own Salary	✓	✓	✓	✓
View All Payroll	X	✓	✓	✓
Calculate Payroll	X	✓	✓	X
Release Funds	X	X	✓	X
Mark Paid	X	✓	✓	✓
Submit Loan	✓	✓	X	X
Approve Loan	X	X	✓	X
Submit Medical	✓	✓	X	X
Approve Medical	X	X	✓	X
Disburse Medical	X	✓	✓	X
Submit Expense	✓	✓	X	X
Approve Expense	X	✓	✓	X

Business Rules

ID	Rule
BR-PAY-01	net_salary = base_salary + overtime_pay + bonuses + performance_bonus - penalties - late_deductions - absence_deductions
BR-PAY-02	Payroll must be in 'funds_released' status before marking as 'paid'
BR-PAY-03	Only Owner can release funds
BR-PAY-04	Employee can only view own payroll records
BR-PAY-05	monthly_deduction = amount / installments
BR-PAY-06	Loans require Owner approval
BR-PAY-07	Employee must confirm receipt before HR final approval
BR-PAY-08	Deferred months (TEXT[]) skip payroll deduction for that month
BR-PAY-09	Loan auto-completes when remaining_amount reaches 0
BR-PAY-10	Medical requests require Owner approval before HR can disburse

ID	Rule
BR-PAY-11	Payroll status flow: processing → funds_released → paid

PART 4 — SECTION 4: Finance & Accounting

4.1 Section Purpose

- Comprehensive double-entry accounting system managing all financial operations of the factory: treasury, banking, invoicing, check lifecycle, journal entries, cost centers, fixed assets, taxes, and financial closings.
- **Why:** A manufacturing operation requires rigorous financial tracking across multiple revenue streams, supplier payments, payroll, and maintenance costs
 - **Problem Solved:** Integrated GL with full invoice lifecycle, check management, custody tracking, tax filing, and period closings
 - **Users:** CFO, Chief Accountant, Accountant, Owner

4.2 Actors

Actor	Responsibilities
CFO	Strategic financial oversight, Approve large transactions, Financial reporting
Chief Accountant	Journal entry approval, Financial closing, Tax filing
Accountant	Data entry (invoices, payments, checks), Daily operations
Owner	Final approval for major financial decisions
Procurement Manager	Purchase invoice creation
Sales Manager	Sales invoice creation

4.3 Features

Feature	Actor	Tables	CRUD
Manage Safes (Treasury)	Accountant	finance_safes	C/R/U
Manage Bank Accounts	Accountant	finance_bank_accounts	C/R/U
Manage Finance Clients	Sales/Accountant	finance_clients	C/R/U
Manage Finance Suppliers	Procurement/Accountant	finance_suppliers	C/R/U
Create Invoice (Sales/Purchase/Return)	Accountant	finance_invoices	C/R/U
Record Collection/Payment	Accountant	finance_collections	C/R
Manage Checks (Incoming/Outgoing)	Accountant	finance_checks	C/R/U

Feature	Actor	Tables	CRUD
Issue Custody/Advance	Accountant	finance_custodies	C/R/U
Settle Custody	Employee/Accountant	finance_custodies	U
Create Journal Entry	Accountant	finance_journal_entries, finance_journal_lines	C/R
Approve Journal Entry	Chief Accountant	finance_journal_entries	U
Post Journal Entry	Chief Accountant	finance_journal_entries	U
Manage Cost Centers	CFO	finance_cost_centers	C/R/U
Manage Fixed Assets	Accountant	finance_fixed_assets	C/R/U
Manage Expense Categories	CFO	finance_expense_categories	C/R/U
Tax Calculation & Filing	Chief Accountant	finance_taxes	C/R/U
Financial Closing (Daily/Monthly/Yearly)	Chief Accountant	finance_closings	C/R
Record GL Transaction	System/Accountant	general_ledger	C/R
Manage Chart of Accounts	CFO	chart_of_accounts	C/R/U
Manage Currencies	CFO	currencies	C/R/U
Update Exchange Rates	Accountant	exchange_rates	C/R
Department Budgets	CFO	department_budgets	C/R/U
Auto Document Numbering	System	doc_sequences	R/U

Processes

Process FIN-P01: Invoice Lifecycle

flowchart TD

```

A[👤 Accountant] --> B[Create Invoice]
B --> C{Invoice Type}
C -->|Sales| D[Link to finance_client]
C -->|Purchase| E[Link to finance_supplier]
D --> F[Calculate: subtotal - discount + tax + additions = total]
E --> F
F --> G[Status: unpaid]
G --> H[Record Payment/Collection]
H --> I{Full Payment?}
I -->|Yes| J[Status: paid]
I -->|No| K[Status: partial]
K --> L[remaining = total - paid_amount]
L --> H
J --> M[Auto GL Entry]
M --> N[👤 Accountant - Complete]

```

Process FIN-P02: Check Lifecycle (Incoming)

flowchart TD

```

A[👤 Receive Check from Client] --> B[Status: pending]
B --> C[Deposit to Bank]
C --> D[Status: under_collection]
D --> E[Bank Result]
E -->|Cleared| F[Status: collected]
E -->|Returned| G[Status: bounced]
F --> H[Update Bank Balance]
G --> I[Update Client Balance]
H --> J[GL Entry: Debit Bank, Credit AR]
I --> K[Notify Sales Manager]

```

Process FIN-P03: Journal Entry Flow

flowchart TD

```

A[👤 Accountant] --> B[Create Journal Entry]
B --> C[Add Journal Lines]
C --> D{Debit = Credit?}
D -->|No| E[❌ Error: Unbalanced]
D -->|Yes| F[Status: draft]
F --> G[Submit for Approval]
G --> H{Chief Accountant Review}
H -->|Approve| I[Status: approved]
H -->|Reject| J[Status: cancelled]
I --> K[Post to GL]
K --> L[Status: posted]
L --> M[Update Account Balances]

```

Process FIN-P04: Financial Closing

flowchart TD

```

A[👤 Chief Accountant] --> B[Select Closing Type]
B --> C{Type}
C -->|Daily| D[Reconcile Today's Transactions]
C -->|Monthly| E[Reconcile Month's Transactions]
C -->|Yearly| F[Full Year Reconciliation]
D --> G[Create Closing Record]
E --> G
F --> G
G --> H[is_locked = true]
H --> I[No Edits Allowed for Closed Period]
I --> J[👤 Chief Accountant - Confirmed]

```

Section ERD

```

erDiagram
    FINANCE_SAFES ||--o{ FINANCE_COLLECTIONS : "receives"
    FINANCE_BANK_ACCOUNTS ||--o{ FINANCE_COLLECTIONS : "receives"
    FINANCE_CLIENTS ||--o{ FINANCE_INVOICES : "billed"
    FINANCE_SUPPLIERS ||--o{ FINANCE_INVOICES : "bills"
    FINANCE_INVOICES ||--o{ FINANCE_COLLECTIONS : "paid_via"
    FINANCE_INVOICES ||--o{ FINANCE_CHECKS : "paid_by"
    FINANCE_CLIENTS ||--o{ FINANCE_CHECKS : "issues"
    FINANCE_SUPPLIERS ||--o{ FINANCE_CHECKS : "receives"
    FINANCE_JOURNAL_ENTRIES ||--|{ FINANCE_JOURNAL_LINES : "contains"
    FINANCE_COST_CENTERS ||--o{ FINANCE_COST_CENTERS : "parent"
    CHART_OF_ACCOUNTS ||--o{ CHART_OF_ACCOUNTS : "parent"

    FINANCE_SAFES {
        uuid id PK
        text name
        numeric balance
        boolean is_active
    }

    FINANCE_BANK_ACCOUNTS {
        uuid id PK
        text bank_name
        text account_number
        numeric balance
        boolean is_active
    }

    FINANCE_CLIENTS {
        uuid id PK
        text name
        text phone
        text email
        numeric credit_limit
        numeric balance
    }

    FINANCE_SUPPLIERS {
        uuid id PK
        text name
        text phone
        numeric balance
    }

    FINANCE_INVOICES {
        uuid id PK
        text invoice_number
        text invoice_type
        uuid client_id FK
        uuid supplier_id FK
        numeric subtotal
        numeric discount
        numeric tax_amount
        numeric total
        numeric paid_amount
        numeric remaining
        text status
    }

```

```
FINANCE_COLLECTIONS {
    uuid id PK
    uuid invoice_id FK
    text collection_type
    numeric amount
    uuid safe_id FK
    uuid bank_id FK
    text reference_number
}

FINANCE_CHECKS {
    uuid id PK
    text check_number
    text check_type
    numeric amount
    text bank_name
    date due_date
    text status
    uuid client_id FK
    uuid supplier_id FK
    uuid invoice_id FK
}

FINANCE_CUSTODIES {
    uuid id PK
    uuid employee_id
    text employee_name
    numeric amount
    numeric settled_amount
    numeric remaining
    text status
}

FINANCE_JOURNAL_ENTRIES {
    uuid id PK
    text entry_number
    date entry_date
    text description
    numeric total_debit
    numeric total_credit
    text status
    uuid approved_by
}

FINANCE_JOURNAL_LINES {
    uuid id PK
    uuid entry_id FK
    text account_name
    numeric debit
    numeric credit
    text cost_center
}

FINANCE_COST_CENTERS {
    uuid id PK
    text name
    text code
    uuid parent_id FK
    boolean is_active
}
```

```
FINANCE_FIXED_ASSETS {  
    uuid id PK  
    text name  
    text asset_code  
    text category  
    date purchase_date  
    numeric purchase_cost  
    numeric depreciation_rate  
    numeric accumulated_depreciation  
    numeric current_value  
    text status  
}
```

```
FINANCE_TAXES {  
    uuid id PK  
    text tax_type  
    text period  
    numeric taxable_amount  
    numeric tax_amount  
    text status  
}
```

```
FINANCE_CLOSINGS {  
    uuid id PK  
    text closing_type  
    text period  
    boolean is_locked  
}
```

```
CHART_OF_ACCOUNTS {  
    uuid id PK  
    text code UK  
    text name_en  
    text name_ar  
    text account_type  
    uuid parent_id FK  
    numeric balance  
}
```

```
GENERAL_LEDGER {  
    uuid id PK  
    timestampz transaction_date  
    text account_type  
    text account_name  
    numeric debit  
    numeric credit  
    text reference_type  
    uuid reference_id  
}
```

```
DEPARTMENT_BUDGETS {  
    uuid id PK  
    text department  
    text fiscal_year  
    text budget_type  
    numeric allocated_amount  
    numeric spent_amount  
    numeric remaining_amount  
    text status  
}
```

```
DOC_SEQUENCES {
```

```
    uuid id PK
    text doc_type UK
    text prefix
    integer last_number
}
```

Business Rules

ID	Rule
BR-FIN-01	Invoice total = subtotal - discount + tax_amount + additions
BR-FIN-02	remaining = total - paid_amount
BR-FIN-03	Invoice status: unpaid → partial → paid (based on remaining)
BR-FIN-04	Check lifecycle: pending → under_collection → collected/bounced/cancelled
BR-FIN-05	Journal entry must balance: total_debit = total_credit
BR-FIN-06	Journal workflow: draft → approved → posted / cancelled
BR-FIN-07	Posted journal entries cannot be edited
BR-FIN-08	Financial closing locks period — no edits allowed
BR-FIN-09	Cost centers support hierarchy (parent_id self-reference)
BR-FIN-10	Chart of accounts supports hierarchy (parent_id self-reference)
BR-FIN-11	Fixed asset current_value = purchase_cost - accumulated_depreciation
BR-FIN-12	Custody remaining = amount - settled_amount
BR-FIN-13	Document numbers auto-increment via doc_sequences
BR-FIN-14	Budget remaining = allocated_amount - spent_amount
BR-FIN-15	Each (department, fiscal_year, budget_type) is unique

Permissions Matrix

Feature	Accountant	Chief Acct	CFO	Owner
Manage Safes	✓	✓	✓	✓
Manage Bank Accounts	✓	✓	✓	✓
Create Invoices	✓	✓	✓	✓
Record Payments	✓	✓	✓	✓
Manage Checks	✓	✓	✓	✓
Create Journal Entry	✓	✓	✓	✓
Approve Journal Entry	X	✓	✓	✓
Post Journal Entry	X	✓	✓	✓
Financial Closing	X	✓	✓	✓
Manage Chart of Accounts	X	✓	✓	✓

Feature	Accountant	Chief Acct	CFO	Owner
Department Budgets	X	X	✓	✓
Tax Filing	X	✓	✓	✓
View Financial Reports	✓	✓	✓	✓

PART 4 — SECTIONS 5-8: Supply Chain & Operations

SECTION 5: Procurement & Purchasing

5.1 Purpose

Manages the complete procurement lifecycle from departmental purchase requests through supplier quotations, purchase order creation, goods receipt, and settlement. Includes AI-powered supplier performance evaluation.

5.2 Actors

Actor	Responsibilities
Any Department Head	Submit purchase requests
Warehouse Manager	Check stock availability, dispense items
Procurement Manager	Approve requests, create POs, manage suppliers
Procurement Specialist	Request quotations, source suppliers
Quality Manager	Inspect incoming materials
Owner	Approve major purchases

5.3 Purchase Request Workflow

Department → pending_warehouse → Warehouse checks stock →
IF available: dispensed (from warehouse)
IF not available: pending_manager → Procurement Manager approves →
quotation_requested → purchased → pending_finance → received

5.4 Features

Feature	Actor	Tables	CRUD
Submit Purchase Request	Dept Head	purchase_requests	C
Warehouse Stock Check	Warehouse Mgr	purchase_requests, inventory_items	R/U
Dispense from Stock	Warehouse Mgr	purchase_requests, inventory_transactions	U/C
Approve Purchase Request	Procurement Mgr	purchase_requests	U

Feature	Actor	Tables	CRUD
Create Purchase Order	Procurement Specialist	purchase_orders	C
Approve Purchase Order	Procurement Mgr / Owner	purchase_orders	U
Receive Goods	Warehouse Mgr	purchase_orders, inventory_items	U
QC Incoming Material	QC Inspector	qc_inspections, purchase_orders	C/U
Manage Suppliers	Procurement Mgr	suppliers	C/R/U/D
Supplier Portal Login	Supplier	suppliers	R
View Supplier Orders	Supplier	supplier_orders	R
Record Supplier Transaction	Accountant	supplier_transactions	C
AI Supplier Performance	AI/Procurement	supplier_performance	C/R

Process Flow

```
flowchart TD
    A[👤 Department Head] --> B[Submit Purchase Request]
    B --> C[Status: pending_warehouse]
    C --> D[Warehouse Check]
    D -->|In Stock| E[Dispense from Warehouse]
    D -->|Out of Stock| F[Status: pending_manager]
    E --> G[Status: dispensed]
    F --> H[Procurement Manager]
    H -->|Approve| I[Status: approved]
    H -->|Reject| J[Status: rejected]
    I --> K[Request Quotations from Suppliers]
    K --> L[Status: quotation_requested]
    L --> M[Create Purchase Order]
    M --> N[Status: purchased]
    N --> O[Goods Arrive at Warehouse]
    O --> P[QC Inspection]
    P --> Q[QC Result]
    Q -->|Pass| R[Status: received]
    Q -->|Fail| S[Return to Supplier]
    R --> T[Update Inventory]
    T --> U[Status: pending_finance]
    U --> V[Finance Settlement]
    V --> W[👤 Department Head - Complete]
```

Section ERD

```

erDiagram
    PURCHASE_REQUESTS ||--o{ PURCHASE_ORDERS : "fulfilled_by"
    PURCHASE_REQUESTS }o--|| INVENTORY_ITEMS : "for_item"
    SUPPLIERS ||--o{ SUPPLIER_ORDERS : "receives"
    SUPPLIERS ||--o{ SUPPLIER_TRANSACTIONS : "paid_via"
    SUPPLIERS ||--o{ SUPPLIER_PERFORMANCE : "evaluated"
    SUPPLIERS ||--o{ PURCHASE_ORDERS : "supplies"

    PURCHASE_REQUESTS {
        uuid id PK
        uuid item_id FK
        text item_name
        integer requested_quantity
        text unit
        date delivery_date
        text status
        text requested_by
        text department
        text approved_by
    }

    PURCHASE_ORDERS {
        uuid id PK
        text po_number
        uuid supplier_id FK
        text supplier_name
        date order_date
        date delivery_date
        jsonb items
        numeric total
        text status
        text quality_status
        uuid approved_by FK
    }

    SUPPLIERS {
        uuid id PK
        text company_name
        text contact_person
        text email UK
        text phone
        text password_hash
    }

    SUPPLIER_ORDERS {
        uuid id PK
        uuid supplier_id FK
        text order_details
        numeric total_amount
        text status
    }

    SUPPLIER_TRANSACTIONS {
        uuid id PK
        uuid supplier_id FK
        numeric amount
        text payment_method
        text check_status
  
```

```
    date transaction_date
  }

  SUPPLIER_PERFORMANCE {
    uuid id PK
    uuid supplier_id FK
    date period_start
    date period_end
    integer total_orders
    integer on_time_deliveries
    integer late_deliveries
    numeric quality_score
    numeric overall_score
    text ai_recommendation
  }
```

Business Rules

ID	Rule
BR-PROC-01	PR workflow: pending_warehouse → dispensed OR pending_manager → approved → quotation_requested → purchased → pending_finance → received
BR-PROC-02	Warehouse checks stock before escalating to procurement
BR-PROC-03	PO must reference a supplier
BR-PROC-04	PO quality_status must be 'approved' before closing
BR-PROC-05	Supplier performance AI score considers: on-time delivery, quality, price competitiveness, return rate
BR-PROC-06	Suppliers can login to portal with email + password_hash

SECTION 6: Inventory & Warehouse

6.1 Purpose

Real-time inventory tracking with stock movements, low-stock alerts, and material traceability across all warehouses.

6.2 Actors

Actor	Responsibilities
Warehouse Manager	Full CRUD on inventory, Process movements
Production Manager	Request materials (out), Return excess
Procurement Manager	Add incoming stock (in)

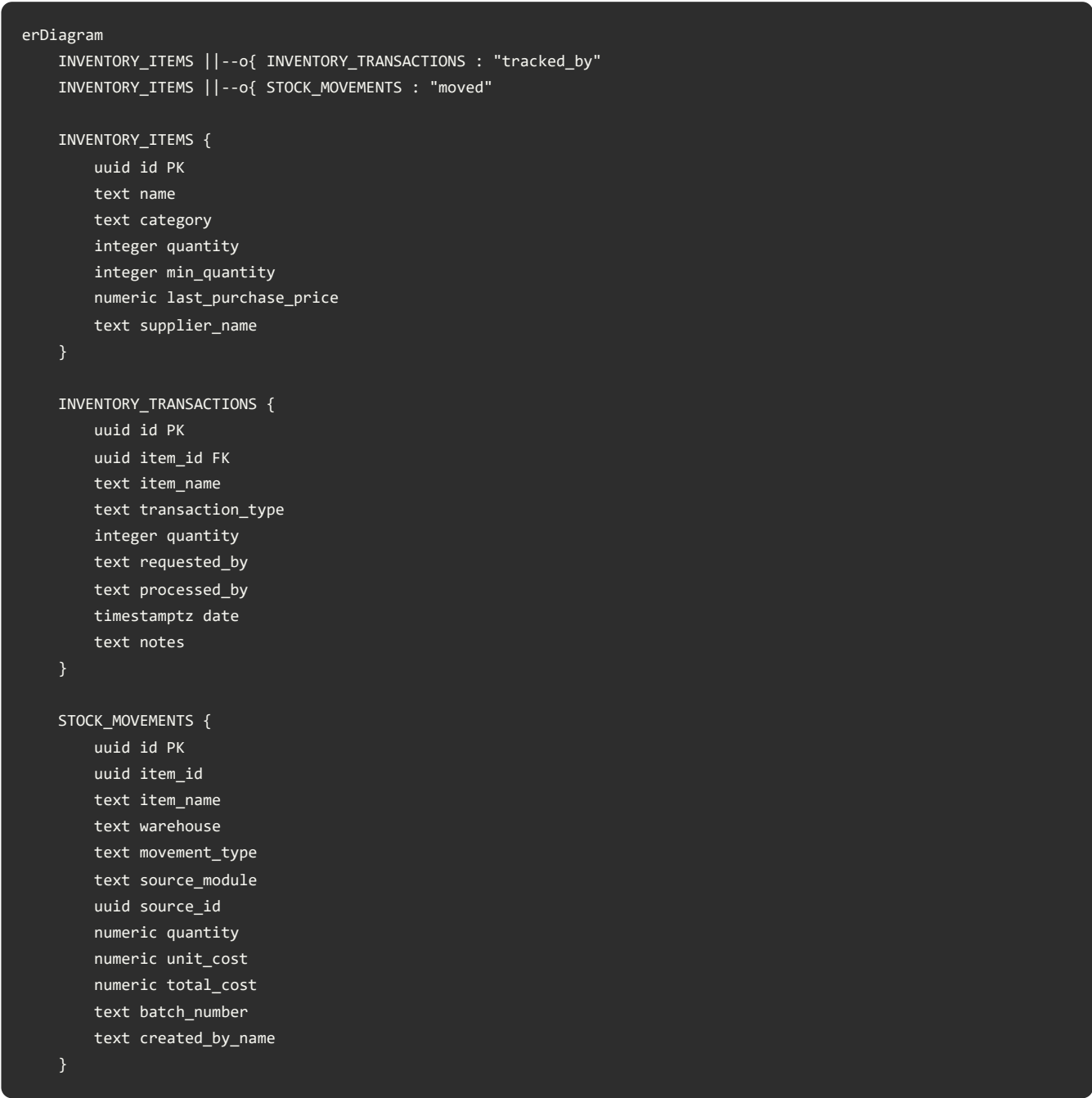
6.3 Features

Feature	Actor	Tables	CRUD
Add Inventory Item	Warehouse Mgr	inventory_items	C
Update Item	Warehouse Mgr	inventory_items	U
Stock In (Receive)	Warehouse Mgr	inventory_transactions, stock_movements	C
Stock Out (Dispense)	Warehouse Mgr	inventory_transactions, stock_movements	C
Stock Transfer	Warehouse Mgr	stock_movements	C
Stock Adjustment	Warehouse Mgr	stock_movements	C
Low Stock Alert	System	inventory_items (quantity < min_quantity)	R
View Stock Report	Warehouse Mgr	inventory_items, stock_movements	R

Process Flow

```
flowchart TD
    A[📦 Stock Movement Event] --> B{Movement Type}
    B -->|In| C[Increase quantity]
    B -->|Out| D[Decrease quantity]
    B -->|Transfer| E[Move between warehouses]
    B -->|Adjustment| F[Correct quantity]
    B -->|Return| G[Return to stock]
    C --> H[Update inventory_items.quantity]
    D --> H
    E --> H
    F --> H
    G --> H
    H --> I{quantity < min_quantity?}
    I -->|Yes| J[🚨 Low Stock Alert]
    I -->|No| K[Normal]
    J --> L[Notify Warehouse Manager]
```

Section ERD



Business Rules

ID	Rule
BR-INV-01	Categories: Maintenance, Workshop, Supplies, Chemicals, Fixed Assets
BR-INV-02	Low stock alert when quantity < min_quantity
BR-INV-03	Transaction types: in, out
BR-INV-04	Movement types: in, out, transfer, adjustment, return
BR-INV-05	Stock out cannot reduce quantity below 0
BR-INV-06	Movements linked to source_module (procurement, production, sales)

SECTION 7: Production & Manufacturing

7.1 Purpose

Manages manufacturing operations from Bill of Materials through production stages to finished batches with full material traceability.

7.2 Actors

Actor	Responsibilities
Production Manager	Create/manage production orders, BOM, stages
Planning Manager	Production planning and scheduling
Quality Manager	QC at production stages
Warehouse Manager	Provide raw materials

7.3 Features

Feature	Tables	CRUD
Manage BOM	bom, bom_items	C/R/U/D
Create Production Order	production_orders	C
Define Stage Templates	production_stage_templates	C/R/U
Track Production Stages	production_stage_logs	C/R/U
Record Material Consumption	production_consumption	C/R
Create Batch	production_batches	C
Track Batch Quality	production_batches, qc_inspections	R/U

Process Flow

flowchart TD

```
A[👤 Production Manager] --> B[Create Production Order]
B --> C[Select Product + BOM]
C --> D[Request Materials from Warehouse]
D --> E[Record Consumption]
E --> F[Stage 1: Raw Material Prep]
F --> G{QC Required?}
G -->|Yes| H[QC Inspection]
G -->|No| I[Next Stage]
H -->|Pass| I
H -->|Fail| J[Stage Failed - Rework]
I --> K[Stage 2: Processing]
K --> L[Stage N: Final]
L --> M[Create Batch]
M --> N[Batch QC Inspection]
N -->|Pass| O[Status: finished]
N -->|Fail| P[Status: qc_failed]
O --> Q[Move to Finished Goods Warehouse]
Q --> R[👤 Warehouse Manager]
```

Start Actor: Production Manager | **End Actor:** Warehouse Manager (receives finished goods)

Section ERD

```

erDiagram
    PRODUCTS ||--o{ BOM : "has"
    BOM ||--|{ BOM_ITEMS : "contains"
    BOM_ITEMS }o--|| INVENTORY_ITEMS : "uses"
    PRODUCTION_ORDERS ||--o{ PRODUCTION_STAGE_LOGS : "has"
    PRODUCTION_ORDERS ||--o{ PRODUCTION_CONSUMPTION : "consumes"
    PRODUCTION_ORDERS ||--o{ PRODUCTION_BATCHES : "produces"
    PRODUCTION_STAGE_TEMPLATES ||--o{ PRODUCTION_STAGE_LOGS : "defines"
    PRODUCTION_BATCHES }o--o| QC_INSPECTIONS : "inspected_by"
    PRODUCTS ||--o{ PRODUCTION_BATCHES : "produced_as"

    BOM {
        uuid id PK
        uuid product_id FK
        text name
        integer version
        text status
    }

    BOM_ITEMS {
        uuid id PK
        uuid bom_id FK
        text material_name
        uuid material_id FK
        numeric quantity
        text unit
        numeric waste_percent
        numeric cost_per_unit
    }

    PRODUCTION_STAGE_LOGS {
        uuid id PK
        uuid production_order_id FK
        uuid stage_template_id FK
        text stage_name
        text status
        timestamptz started_at
        timestamptz completed_at
        numeric quantity_in
        numeric quantity_out
        numeric waste_quantity
        text qc_result
    }

    PRODUCTION_CONSUMPTION {
        uuid id PK
        uuid production_order_id FK
        uuid material_id FK
        numeric planned_qty
        numeric actual_qty
        numeric waste_qty
        uuid batch_id
    }

    PRODUCTION_BATCHES {
        uuid id PK
        text batch_number UK
        uuid production_order_id FK

```

```
    uuid product_id FK
    numeric quantity
    text status
    uuid qc_inspection_id FK
    date manufactured_date
    date expiry_date
    text warehouse_location
}
```

Business Rules

ID	Rule
BR-PROD-01	BOM versions allow multiple recipes per product
BR-PROD-02	Stage workflow: pending → in_progress → completed/failed/skipped
BR-PROD-03	Stages with is_qc_required=true must pass QC before proceeding
BR-PROD-04	Material consumption tracks planned vs actual vs waste
BR-PROD-05	Batch lifecycle: in_production → qc_pending → qc_passed/qc_failed → finished → shipped
BR-PROD-06	Traceability: Batch → Production Order → BOM → Materials → Supplier
BR-PROD-07	waste_quantity = quantity_in - quantity_out (per stage)

SECTION 8: Quality Control

8.1 Purpose

Quality gate for both incoming raw materials and finished products, ensuring factory output meets standards.

8.2 Actors

Actor	Responsibilities
QC Inspector	Perform inspections, Record results
Quality Manager	Review inspections, Approve conditional items
Production Manager	Submit batches for QC
Procurement Manager	Submit incoming materials for QC

8.3 Features

Feature	Tables	CRUD
Create Incoming Material Inspection	qc_inspections	C
Create Finished Product Inspection	qc_inspections	C
Record Inspection Result	qc_inspections	U
Pass/Fail Decision	qc_inspections	U

Feature	Tables	CRUD
Conditional Approval	qc_inspections	U
View QC Dashboard	qc_inspections	R

Process Flow

```
flowchart TD
    A[📦 Material/Product Arrives] --> B{Type}
    B -->|Incoming Material| C[QC Inspector Inspects]
    B -->|Finished Product| D[QC Inspector Inspects Batch]
    C --> E{Result}
    D --> E
    E -->|Passed| F[Status: passed]
    E -->|Failed| G[Status: failed]
    E -->|Conditional| H[Status: conditional_approval]
    F --> I[Accept into Inventory / Ship]
    G --> J[Return / Rework / Scrap]
    H --> K[Quality Manager Review]
    K --> L{Decision}
    L -->|Accept| I
    L -->|Reject| J
```

Section ERD

```
erDiagram
    QC_INSPECTIONS {
        uuid id PK
        text reference_type
        uuid reference_id
        text item_name
        text inspector_name
        text status
        text notes
        timestamptz inspection_date
    }
```

Business Rules

ID	Rule
BR-QC-01	reference_type: incoming_material or finished_product
BR-QC-02	Status: pending → passed / failed / conditional_approval
BR-QC-03	Failed incoming materials must be returned to supplier
BR-QC-04	Failed batches must be reworked or scrapped
BR-QC-05	QC inspections linked to batches via production_batches.qc_inspection_id

Cross-Section Relationships (5-6-7-8)

```
flowchart LR
    PROC[📄 Procurement] -->|Purchase Order| QC1[🔍 QC: Incoming]
    QC1 -->|Passed| INV[📦 Inventory]
    INV -->|Raw Materials| PROD[🏭 Production]
    PROD -->|Finished Batch| QC2[🔍 QC: Finished]
    QC2 -->|Passed| INV2[📦 Finished Goods]
    INV2 -->|Ready to Ship| SALES[📊 Sales]
```

PART 4 — SECTIONS 9-12: Customer-Facing & Operations

SECTION 9: Sales & CRM 📊

9.1 Purpose

End-to-end sales management from public product catalog through quote requests, customer registration, sales order processing, and delivery. The public-facing catalog allows guests to browse products and request quotes without authentication.

9.2 Actors

Actor	Responsibilities
Sales Manager	Manage clients, process sales orders, manage products
Customer/Guest	Browse catalog, register, request quotes
Owner	Approve large orders, view sales dashboard
Logistics Manager	Delivery of confirmed orders

9.3 Process Flows

Sales Order Lifecycle

```

flowchart TD
    A[👤 Customer] --> B[Browse Public Catalog]
    B --> C[Request Quote]
    C --> D[👤 Sales Manager Reviews]
    D --> E{Existing Client?}
    E -->|No| F[Customer Registration Request]
    F --> G{Sales Approves Registration?}
    G -->|Yes| H[Create Client Record]
    G -->|No| I[Reject]
    E -->|Yes| J[Create Sales Order]
    H --> J
    J --> K[Status: draft]
    K --> L[Confirm Order]
    L --> M[Status: confirmed]
    M --> N[Production / Stock Check]
    N --> O[Status: delivered]
    O --> P[Create Invoice]
    P --> Q[Status: invoiced]
    Q --> R[Payment Received]
    R --> S[Status: paid]
    S --> T[👤 Customer - Complete]
  
```

Start Actor: Customer | **End Actor:** Customer (receives product)

Quote Request Flow

```

flowchart TD
    A[👤 Guest/Customer] --> B[Fill Quote Form]
    B --> C[Select Product + Quantity + Specs]
    C --> D[Submit Quote Request]
    D --> E[Status: pending]
    E --> F[👤 Sales Manager Reviews]
    F --> G{Decision}
    G -->|Quote| H[Status: quoted - Send Price]
    G -->|Reject| I[Status: rejected]
    H --> J{Customer Accepts?}
    J -->|Yes| K[Convert to Sales Order]
    K --> L[Status: converted_to_so]
    J -->|No| M[No Action]
  
```

Section ERD

```

erDiagram
    PRODUCT_CATEGORIES ||--o{ PRODUCTS : "contains"
    PRODUCTS ||--o{ PRODUCT_IMAGES : "has"
    PRODUCTS ||--o{ PRODUCT_DOCUMENTS : "has"
    PRODUCTS ||--o{ QUOTE_REQUESTS : "quoted_for"
    CLIENTS ||--o{ SALES_ORDERS : "places"
    CLIENTS ||--o{ QUOTE_REQUESTS : "requests"
    PRODUCTS ||--o{ CUSTOMER_REGISTRATION_REQUESTS : "interested_in"
    CLIENTS ||--o{ CUSTOMER_FEEDBACK : "gives"

    PRODUCT_CATEGORIES {
        uuid id PK
        text name_ar
        text name_en
        text slug UK
        text icon
    }

    PRODUCTS {
        uuid id PK
        text name_ar
        text name_en
        text slug UK
        text code
        text sku
        uuid category_id FK
        text material
        text color
        text grade
        jsonb technical_specs
        text unit
        integer moq
        boolean is_public
        boolean is_featured
        numeric cost
        numeric selling_price
        integer current_stock
        integer views
    }

    PRODUCT_IMAGES {
        uuid id PK
        uuid product_id FK
        text image_url
        boolean is_main
        integer display_order
    }

    PRODUCT_DOCUMENTS {
        uuid id PK
        uuid product_id FK
        text title_ar
        text file_url
        text doc_type
        boolean is_public
    }

    SALES_ORDERS {

```

```
    uuid id PK
    text so_number
    uuid client_id FK
    text client_name
    date order_date
    jsonb items
    numeric total
    numeric paid_amount
    numeric remaining
    text invoice_type
    text status
    text delivery_status
}

CLIENTS {
    uuid id PK
    text company_name
    text contact_person
    text phone
    text email
    text address
    text status
}

QUOTE_REQUESTS {
    uuid id PK
    text quote_number UK
    uuid product_id FK
    uuid client_id FK
    text company_name
    numeric quantity
    text required_specs
    date required_delivery_date
    text status
}

CUSTOMER_REGISTRATION_REQUESTS {
    uuid id PK
    text company_name
    text contact_person
    text phone
    uuid requested_product_id FK
    text status
}

CUSTOMER_FEEDBACK {
    uuid id PK
    text customer_name
    uuid order_id
    integer rating
    text feedback_type
    text message
    text status
    text response
}
```

Business Rules

ID	Rule
BR-SALES-01	SO workflow: draft → confirmed → delivered → invoiced → paid / cancelled
BR-SALES-02	Invoice types: cash, credit, tax
BR-SALES-03	remaining = total - paid_amount
BR-SALES-04	Public catalog shows only products where is_public=true AND status='active'
BR-SALES-05	Quote workflow: pending → reviewed → quoted → rejected / converted_to_so
BR-SALES-06	Customer registration requires Sales Manager approval
BR-SALES-07	Product views counter incremented on public page view
BR-SALES-08	Feedback rating: 1-5 stars

SECTION 10: Fleet & Logistics

10.1 Purpose

Comprehensive fleet management including internal/external drivers, company vehicles with odometer tracking, trip scheduling, incident management, and fleet maintenance.

10.2 Actors

Actor	Responsibilities
Logistics Manager	Full fleet management, trip scheduling
Internal Driver	Trip execution, odometer recording
External Driver	External transport services
Owner	View fleet dashboard, costs

10.3 Process Flows

Trip Lifecycle

```
flowchart TD
    A[👤 Logistics Manager] --> B[Create Trip]
    B --> C{Driver Type}
    C -->|Internal| D[Assign Internal Driver + Vehicle]
    C -->|External| E[Assign External Driver]
    D --> F[Status: مجدولة - Scheduled]
    E --> F
    F --> G[Loading]
    G --> H[Status: جارى التحميل]
    H --> I[Departure]
    I --> J[Status: في الطريق]
    J --> K[Arrival]
    K --> L[Status: تم التوصيل]
    L --> M[Record Distance + Fuel]
    M --> N[Calculate Transport Cost]
    N --> O[👤 Logistics Manager - Complete]
```

Fleet Alert System

```
flowchart TD
    A[🔔 System Daily Check] --> B{Check Alerts}
    B --> C[Driver License < 30 days]
    B --> D[Vehicle License < 30 days]
    B --> E[Insurance < 30 days]
    B --> F[Odometer near Maintenance Due]
    C --> G[🚨 Alert: License Expiry]
    D --> G
    E --> G
    F --> H[🚨 Alert: Maintenance Due]
    G --> I[Notify Logistics Manager]
    H --> I
```

Section ERD

erDiagram

```

FLEET_INTERNAL_DRIVERS ||--o{ FLEET_TRIPS : "drives"
FLEET_EXTERNAL_DRIVERS ||--o{ FLEET_TRIPS : "drives"
FLEET_VEHICLES ||--o{ FLEET_TRIPS : "used_in"
FLEET_VEHICLES ||--o{ FLEET_MAINTENANCE : "maintained"
FLEET_VEHICLES ||--o{ FLEET_INCIDENTS : "involved_in"
FLEET_TRIPS ||--o{ FLEET_INCIDENTS : "has"
FLEET_INTERNAL_DRIVERS ||--o| FLEET_VEHICLES : "assigned_to"
CARS ||--o{ CAR_TRIPS : "has"
CARS ||--o{ CAR_MAINTENANCE_LOGS : "maintained"

```

```

FLEET_INTERNAL_DRIVERS {
    uuid id PK
    uuid user_id FK
    text driver_code UK
    text driver_name
    text license_number
    date license_expiry
    text status
    integer trips_count
    numeric total_km
    numeric rating
}

```

```

FLEET_EXTERNAL_DRIVERS {
    uuid id PK
    text driver_name
    text company_name
    text car_number
    numeric capacity
    numeric transport_rate
    text status
}

```

```

FLEET_VEHICLES {
    uuid id PK
    text car_number UK
    text plate_number
    text model
    integer year
    uuid current_driver_id FK
    text status
    numeric odometer
    numeric fuel_consumption
    date insurance_expiry
    date license_expiry
}

```

```

FLEET_TRIPS {
    uuid id PK
    text trip_number UK
    text client_name
    uuid sales_order_id
    text driver_type
    uuid internal_driver_id FK
    uuid external_driver_id FK
    uuid vehicle_id FK
    text status
}

```

```
        numeric distance
        numeric transport_cost
        numeric fuel_qty
    }

    FLEET_INCIDENTS {
        uuid id PK
        uuid trip_id FK
        uuid vehicle_id FK
        text incident_type
        text details
        numeric cost
    }

    FLEET_MAINTENANCE {
        uuid id PK
        uuid vehicle_id FK
        text maintenance_type
        text details
        numeric cost
        text status
    }

    CARS {
        uuid id PK
        text plate_number UK
        text car_model
        uuid driver_id FK
        numeric current_odometer
        date license_expiry_date
        text status
    }

    CAR_TRIPS {
        uuid id PK
        uuid car_id FK
        uuid driver_id FK
        date trip_date
        text destination
        numeric odometer_start
        numeric odometer_end
        numeric distance_covered
        numeric fuel_consumed_liters
        text status
    }

    CAR_MAINTENANCE_LOGS {
        uuid id PK
        uuid car_id FK
        text log_type
        numeric cost
        numeric odometer_at_log
    }
}
```

Business Rules

ID	Rule
BR-FLEET-01	Trip status (Arabic): ملغاة / تم التوصيل → في الطريق → جاري التحميل → مجدولة

ID	Rule
BR-FLEET-02	Incident types: مخالفة (violation), حادث (accident)
BR-FLEET-03	Fleet alerts: License expiry < 30 days, Insurance expiry < 30 days, Maintenance due
BR-FLEET-04	distance_covered = odometer_end - odometer_start (GENERATED column)
BR-FLEET-05	Car maintenance log types: Fuel, Maintenance, Tires, Oil Change
BR-FLEET-06	Trips linked to sales_orders (delivery) and purchase_orders (pickup)

SECTION 11: Engineering

11.1 Purpose

Manage engineering projects (civil, electrical, mechanical, production line) and technical drawings with multi-department approval workflow.

11.2 Process Flow

```
flowchart TD
    A[👤 Engineering Manager] --> B[Create Project]
    B --> C[Status: planning]
    C --> D[Design Phase]
    D --> E[Upload Drawings]
    E --> F[Status: designing]
    F --> G[Drawing Approval]
    G -->|Approved| H[Status: in_progress]
    G -->|Rejected| I[Revise Drawing]
    I --> E
    H --> J[Construction/Implementation]
    J --> K[Status: supervision]
    K --> L[Project Complete]
    L --> M[Status: completed]
    M --> N[Workflow Approvals by Departments]
    N --> O[👤 Engineering Manager - Complete]
```

Section ERD

```
erDiagram
    USERS ||--o{ ENGINEERING_PROJECTS : "manages"
    ENGINEERING_PROJECTS ||--o{ ENGINEERING_DRAWINGS : "has"
    USERS ||--o{ ENGINEERING_DRAWINGS : "uploads"

    ENGINEERING_PROJECTS {
        uuid id PK
        uuid employee_id FK
        text engineer_name
        text title
        text project_type
        text description
        text technical_specs
        integer progress
        text status
        jsonb workflow_approvals
    }

    ENGINEERING_DRAWINGS {
        uuid id PK
        uuid project_id FK
        uuid uploaded_by FK
        text title
        text file_url
        text status
        text approved_by
        text comments
    }
```

Business Rules

ID	Rule
BR-ENG-01	Project types: civil, electrical, mechanical, production_line, other
BR-ENG-02	Project workflow: planning → designing → in_progress → supervision → completed
BR-ENG-03	Drawing workflow: pending → approved / rejected
BR-ENG-04	workflow_approvals (JSONB) tracks per-department sign-offs
BR-ENG-05	Progress tracked as percentage (0-100)

SECTION 12: Maintenance & Facilities

12.1 Purpose

Machine maintenance tracking, external maintenance company management, equipment registry with rental tracking, and facilities management.

12.2 Process Flows

Maintenance Visit Flow

flowchart TD

```

A[🔧 Issue Reported] --> B[Create Maintenance Request]
B --> C{Internal or External?}
C -->|Internal Technician| D[Assign Technician]
C -->|External Company| E[Create Maintenance Visit]
D --> F[Repair + Log]
E --> G[Company Sends Technician]
G --> H[Work Performed]
H --> I[Record: labor_cost + parts_cost]
I --> J[Before/After Photos]
J --> K[Status: completed]
K --> L{Payment}
L --> M[Create Maintenance Payment]
M --> N{Payment Method}
N -->|Treasury| O[Deduct from Safe]
N -->|Bank| P[Bank Transfer]
O --> Q[Status: paid]
P --> Q
F --> R[Update machine status]
Q --> R
R --> S[👤 Maintenance Manager]
  
```

Equipment Rental Flow

flowchart TD

```

A[👤 Maintenance Mgr] --> B[Need Equipment]
B --> C{Own or Rent?}
C -->|Own| D[Check Equipment Registry]
C -->|Rent| E[Create Rental Contract]
E --> F[Status: draft]
F --> G{Approval}
G -->|Approved| H[Status: active]
H --> I[Equipment in Use]
I --> J{Contract Expiring?}
J -->|< 30 days| K[Status: expiring - Alert]
J -->|Expired| L[Status: expired]
K --> M{Renew or Return?}
L --> M
M -->|Return| N[Status: returned]
M -->|Renew| O[Extend Contract]
  
```

Section ERD

erDiagram

```
MACHINES ||--o{ MAINTENANCE_LOGS : "has"
MAINTENANCE_COMPANIES ||--o{ MAINTENANCE_VISITS : "performs"
MAINTENANCE_VISITS ||--o{ MAINTENANCE_PAYMENTS : "paid_via"
EQUIPMENT_CATEGORIES ||--o{ EQUIPMENT : "categorizes"
EQUIPMENT ||--o{ EQUIPMENT_RENTALS : "rented"
SUPPLIERS ||--o{ EQUIPMENT : "owns"
SUPPLIERS ||--o{ EQUIPMENT_RENTALS : "rents_from"
```

```
MACHINES {
    uuid id PK
    text name
    text code UK
    text location
    text status
    date last_maintenance_date
    date next_maintenance_date
}
```

```
MAINTENANCE_LOGS {
    uuid id PK
    uuid machine_id FK
    text technician_name
    text issue_description
    text action_taken
    text parts_used
    numeric cost
}
```

```
MAINTENANCE_COMPANIES {
    uuid id PK
    text name
    text contact_person
    text phone
    date contract_start
    date contract_end
    numeric rating
    text status
    integer total_visits
    numeric total_spending
}
```

```
MAINTENANCE_VISITS {
    uuid id PK
    text visit_number UK
    uuid company_id FK
    date visit_date
    text problem_description
    text work_description
    numeric labor_cost
    numeric parts_cost
    numeric total_cost
    text status
    text_arr before_photos
    text_arr after_photos
}
```

```
MAINTENANCE_PAYMENTS {
```

```

    uuid id PK
    uuid visit_id FK
    uuid company_id FK
    numeric amount
    text payment_method
    text status
  }

EQUIPMENT {
  uuid id PK
  text name
  text code UK
  uuid category_id FK
  text serial_number
  text status
  uuid owner_supplier_id FK
  numeric daily_rate
  numeric monthly_rate
  numeric purchase_cost
}

EQUIPMENT_RENTALS {
  uuid id PK
  uuid equipment_id FK
  text rental_number UK
  uuid supplier_id FK
  date start_date
  date end_date
  text rate_type
  numeric rate_amount
  numeric total_cost
  text status
}
}
```

Business Rules

ID	Rule
BR-MAINT-01	Machine status: operational, down, under_maintenance
BR-MAINT-02	Visit total_cost = labor_cost + parts_cost
BR-MAINT-03	Visit payment flow: pending → approved → paid / cancelled
BR-MAINT-04	Payment methods: treasury (safe) or bank
BR-MAINT-05	Maintenance company rating: 0-5 scale
BR-MAINT-06	Company status: active, inactive, blacklisted
BR-MAINT-07	Equipment status: available, rented, maintenance, retired
BR-MAINT-08	Rental status: draft → active → expiring → expired / returned / cancelled
BR-MAINT-09	Rate types: daily, weekly, monthly

=====

PART 4 — SECTIONS 13-15 + MASTER ANALYSIS (Parts 24-30)

SECTION 13: Nursing & Factory Clinic

13.1 Purpose

Manages factory clinic operations including employee health visits, work injury documentation (OSHA-style), rest permits, pharmacy inventory, and medication dispensing.

13.2 Actors

Actor	Responsibilities
Nurse	Record clinic visits, injuries, dispense medication, issue rest permits
Nursing Management	Oversee clinic operations, manage pharmacy stock, approve medical leave
HR Manager	View medical reports, integrate with absence/leave system
Owner	View medical dashboards, approve medical reimbursements

13.3 Process Flows

Clinic Visit Flow

```
flowchart TD
    A[👤 Employee] --> B[Visit Factory Clinic]
    B --> C[👩 Nurse Records Visit]
    C --> D[Record Vitals: BP, Temp, HR, Blood Sugar]
    D --> E[Complaint + Diagnosis]
    E --> F{Treatment}
    F -->|Medication| G[Dispense from Pharmacy]
    F -->|Rest| H[Issue Rest Permit]
    F -->|Hospital| I[Hospital Transfer Referral]
    G --> J[Record in clinic_dispense_logs]
    J --> K[Update medication stock]
    H --> L{Permit Type}
    L -->|Clinic Rest| M[Rest in Clinic]
    L -->|Rest of Day| N[Gate Pass Authorized]
    L -->|Sick Leave Short| O[1-3 Day Leave]
    L -->|Sick Leave Extended| P[Extended Medical Leave]
    I --> Q[Record referral_details]
    M --> R{Disposition}
    N --> R
    O --> R
    P --> R
    Q --> R
    R --> S[👤 Employee Returns/Leaves]
```

Work Injury Documentation

flowchart TD

```
A[🚨 Work Injury Occurs] --> B[👩 Nurse Records Injury]
B --> C[Record: location, severity, injury_type]
C --> D[Record: root_cause, immediate_action]
D --> E{Severity}
E -->|Minor/Moderate| F[First Aid + Return to Work]
E -->|Severe/Critical| G[Hospital Transfer]
G --> H[Record: hospital_name, hospitalized=true]
H --> I[Track lost_work_days]
F --> J[Supervisor Notified]
I --> J
J --> K{Investigation Needed?}
K -->|Yes| L[Status: osh_investigation]
K -->|No| M[Status: recovered]
L --> N[Record investigation_notes]
N --> M
```

Section ERD

```

erDiagram
    USERS ||--o{ CLINIC_VISITS : "visits"
    USERS ||--o{ CLINIC_INJURIES : "injured"
    USERS ||--o{ CLINIC_REST_PERMITS : "receives"
    CLINIC_MEDICATIONS ||--o{ CLINIC_DISPENSE_LOGS : "dispensed"
    CLINIC_VISITS ||--o{ CLINIC_DISPENSE_LOGS : "triggers"
    USERS ||--o{ CLINIC_DISPENSE_LOGS : "receives_meds"

    CLINIC_VISITS {
        uuid id PK
        uuid employee_id FK
        text employee_name
        text department
        text visit_type
        text blood_pressure
        numeric temperature
        numeric heart_rate
        numeric blood_sugar
        text complaint
        text diagnosis
        text treatment
        text medicine_dispensed
        text disposition
        text attended_by
    }

    CLINIC_INJURIES {
        uuid id PK
        timestamptz incident_date
        uuid employee_id FK
        text department
        text location
        text severity
        text injury_type
        text description
        text root_cause
        text immediate_action
        boolean hospitalized
        text hospital_name
        numeric lost_work_days
        text status
        text investigation_notes
    }

    CLINIC_REST_PERMITS {
        uuid id PK
        uuid employee_id FK
        text permit_type
        timestamptz start_time
        timestamptz end_time
        text diagnosis
        boolean gate_pass_authorized
        text status
        timestamptz returned_at
        text issued_by
    }

    CLINIC_MEDICATIONS {

```

```
    uuid id PK
    text name
    text generic_name
    text category
    text unit
    numeric current_stock
    numeric min_threshold
    date expiry_date
    text batch_no
    text location
  }

  CLINIC_DISPENSE_LOGS {
    uuid id PK
    uuid medication_id FK
    text medication_name
    uuid employee_id FK
    numeric quantity
    text reason
    uuid visit_id FK
    text dispensed_by
  }
```

Business Rules

ID	Rule
BR-NURSE-01	Visit types: checkup, acute, work_injury, follow_up, first_aid
BR-NURSE-02	Dispositions: return_to_work, clinic_rest, rest_permit, hospital_transfer
BR-NURSE-03	Injury severity: minor, moderate, severe, critical
BR-NURSE-04	Injury types: cut, burn, fracture, chemical, eye, shock, fall, other
BR-NURSE-05	Permit types: clinic_rest, rest_of_day, sick_leave_short, sick_leave_extended
BR-NURSE-06	Medication dispensing decreases current_stock
BR-NURSE-07	Low medication alert when current_stock < min_threshold
BR-NURSE-08	Medication categories: analgesic, antiseptic, wound_care, gastrointestinal, respiratory, burn_care, emergency
BR-NURSE-09	Injury status: under_treatment → rest → hospitalized → recovered / osh_investigation

SECTION 14: Legal & Compliance

14.1 Purpose

Track legal issues, compliance matters, and case management.

14.2 Section ERD

```
erDiagram
    USERS ||--o{ LEGAL_ISSUES : "reports"

    LEGAL_ISSUES {
        uuid id PK
        text title
        text description
        uuid reported_by FK
        text reported_name
        text status
        text assigned_to
    }
```

Business Rules

ID	Rule
BR-LEGAL-01	Default status: 'open'
BR-LEGAL-02	Cases assigned to legal advisor/assigned_to
BR-LEGAL-03	Status flow: open → in_progress → resolved / closed

[!TIP] **Suggestion for Future Enhancement:** This module is minimal. Recommended additions: - Case categories (employment law, contract disputes, regulatory compliance) - Document attachments for legal documents - Deadline tracking for court dates - Linked employees and incidents - Legal cost tracking - External lawyer management

SECTION 15: IT & Administration

15.1 Purpose

Central hub for IT support, system administration, collaboration tools (chat, tasks, calendar), document management, approval workflows, AI-powered reports, and KPI dashboards.

15.2 Subsystems

15.2.1 IT Support

```
erDiagram
    USERS ||--o{ IT_TICKETS : "submits"

    IT_TICKETS {
        uuid id PK
        uuid employee_id FK
        text department
        text issue_type
        text description
        text priority
        text status
        text assigned_to
    }
```

15.2.2 Notifications & Announcements

```
erDiagram
    USERS ||--o{ NOTIFICATIONS : "receives"
    USERS ||--o| NOTIFICATION_PREFERENCES : "configures"

    NOTIFICATIONS {
        uuid id PK
        uuid user_id FK
        text type
        text title
        text message
        boolean read
    }

    ANNOUNCEMENTS {
        uuid id PK
        text title
        text message
        text priority
        text department
    }

    NOTIFICATION_PREFERENCES {
        uuid id PK
        uuid user_id FK
        boolean email_enabled
        boolean push_enabled
        boolean leave_notifications
        boolean attendance_notifications
        boolean payroll_notifications
        boolean task_notifications
        boolean chat_notifications
    }
```

15.2.3 Collaboration (Chat, Tasks, Calendar)

```

erDiagram
    CHAT_CHANNELS ||--|{ CHAT_MESSAGES : "contains"
    USERS ||--o{ CHAT_MESSAGES : "sends"
    USERS ||--o{ TASKS : "assigned"
    USERS ||--o{ CALENDAR_EVENTS : "organizes"

    CHAT_CHANNELS {
        uuid id PK
        text name
        text channel_type
        uuid_arr members
        uuid created_by FK
    }

    CHAT_MESSAGES {
        uuid id PK
        uuid channel_id FK
        uuid sender_id FK
        text message
        text message_type
        text file_url
    }

    TASKS {
        uuid id PK
        text title
        text module
        uuid assigned_to FK
        uuid assigned_by FK
        text priority
        text status
        date due_date
    }

    CALENDAR_EVENTS {
        uuid id PK
        text title
        text event_type
        timestamptz start_time
        timestamptz end_time
        boolean all_day
        uuid organizer_id FK
        uuid_arr attendees
        boolean is_recurring
    }

```

15.2.4 Document Management

```
erDiagram
    DOCUMENT_CATEGORIES ||--o{ DOCUMENTS : "contains"
    DOCUMENT_CATEGORIES ||--o{ DOCUMENT_CATEGORIES : "parent"

    DOCUMENTS {
        uuid id PK
        text title
        uuid category_id FK
        text file_url
        text file_type
        text module
        uuid uploaded_by FK
        integer version
    }

    DOCUMENT_CATEGORIES {
        uuid id PK
        text name
        text name_ar
        uuid parent_id FK
    }
```

15.2.5 Approval Workflows

```
erDiagram
    APPROVAL_WORKFLOWS ||--|{ APPROVAL_STEPS : "has"
    APPROVAL_WORKFLOWS ||--o{ APPROVAL_REQUESTS : "processes"
    APPROVAL_REQUESTS ||--|{ APPROVAL_ACTIONS : "tracks"

    APPROVAL_WORKFLOWS {
        uuid id PK
        text name
        text module
        boolean is_active
    }

    APPROVAL_STEPS {
        uuid id PK
        uuid workflow_id FK
        integer step_order
        text approver_role
        uuid approver_id FK
        text approval_type
        integer auto_approve_after_hours
    }

    APPROVAL_REQUESTS {
        uuid id PK
        uuid workflow_id FK
        integer current_step
        text entity_type
        uuid entity_id
        uuid requested_by FK
        text status
    }

    APPROVAL_ACTIONS {
        uuid id PK
        uuid request_id FK
        integer step_order
        text action
        uuid acted_by FK
        text comments
    }
```

15.2.6 AI Reports & KPIs

```
erDiagram
    USERS ||--o{ AI_REPORTS : "generates"
    KPI_DEFINITIONS ||--o{ KPI_VALUES : "measured_by"

    AI_REPORTS {
        uuid id PK
        text report_type
        text period
        text title
        jsonb content
        text summary
        uuid generated_by FK
        integer report_year
        text language
    }

    KPI_DEFINITIONS {
        uuid id PK
        text name
        text module
        text formula
        text unit
        numeric target_value
        numeric warning_threshold
        numeric danger_threshold
    }

    KPI_VALUES {
        uuid id PK
        uuid kpi_id FK
        text period
        numeric actual_value
        text department
    }
```

PART 24 — CROSS-SECTION RELATIONSHIPS

Inter-Module Data Flow

```
flowchart TD
    subgraph External
        CUST[👤 Customer/Guest]
        SUPP[👤 Supplier]
    end

    subgraph Core
        AUTH[🔒 Auth & Security]
        HR[👥 HR & People]
        PAY[💰 Payroll]
        FIN[🏦 Finance]
    end

    subgraph Supply
        PROC[📋 Procurement]
        INV[📦 Inventory]
    end

    subgraph Manufacturing
        PROD[🏭 Production]
        QC[🔍 Quality]
    end

    subgraph Customer_Facing
        SALES[📊 Sales & CRM]
        FLEET[🚚 Fleet]
    end

    subgraph Support
        ENG[⚙️ Engineering]
        MAINT[🔧 Maintenance]
        NURSE[👩 Nursing]
        LEGAL[⚖️ Legal]
        IT[💻 IT & Admin]
    end

    AUTH -->|Users & Roles| HR
    AUTH -->|Permissions| IT
    HR -->|Employee Data| PAY
    HR -->|Attendance/OT/Deductions| PAY
    PAY -->|Salary Payments| FIN

    CUST -->|Orders| SALES
    SALES -->|Sales Orders| PROD
    SALES -->|Delivery Requests| FLEET
    SALES -->|Sales Invoices| FIN

    SUPP -->|Supply Materials| PROC
    PROC -->|Purchase Orders| INV
    PROC -->|Purchase Invoices| FIN
    PROC -->|Incoming Material| QC
```

```
INV -->|Raw Materials| PROD
PROD -->|Finished Batches| QC
QC -->|Approved Goods| INV

FLEET -->|Driver Employee Data| HR
FLEET -->|Transport Costs| FIN

MAINT -->|Spare Parts| INV
MAINT -->|Payments| FIN
ENG -->|Production Lines| PROD
NURSE -->|Rest Permits| HR
LEGAL -->|Employee Cases| HR

IT -->|Notifications| AUTH
IT -->|AI Reports| FIN
IT -->|Approvals| HR
```

Key Cross-Module Relationships

From Module	To Module	Relationship	Data Shared
HR → Payroll	users.base_salary	Salary base for payroll calculation	
Attendance → Payroll	delay_minutes, overtime_hours	Auto deductions/additions	
Procurement → Inventory	Purchase completed	Stock increase	
Procurement → Quality	Incoming material	QC inspection trigger	
Inventory → Production	Material consumption	Stock decrease	
Production → Quality	Finished batch	Final QC inspection	
Sales → Finance	Sales order invoiced	Revenue recognition	
Sales → Fleet	Delivery order	Trip scheduling	
Procurement → Finance	Purchase invoice	Expense recording	
Fleet → Finance	Transport costs	Cost tracking	
Maintenance → Finance	Maintenance payments	Expense recording	
Maintenance → Inventory	Spare parts used	Stock decrease	
Nursing → HR	Rest permits, injuries	Leave/absence integration	
IT → All	Notifications, Approvals	Cross-cutting concern	

PART 25 — USER / DATA JOURNEYS

Journey 1: Sales Order to Delivery

```

flowchart LR
    A[👤 Customer] -->|Quote Request| B[🏢 Sales]
    B -->|Approved Quote| C[Sales Order Created]
    C -->|Check Stock| D{In Stock?}
    D -->|Yes| E[📦 Reserve from Inventory]
    D -->|No| F[🏭 Create Production Order]
    F -->|Consume Materials| G[📦 Inventory Decrease]
    G -->|Manufacture| H[🏭 Production Stages]
    H -->|Finished Batch| I[🔍 QC Inspection]
    I -->|Passed| J[📦 Add to Finished Goods]
    J -->|Schedule Trip| K[🚚 Schedule Trip]
    K -->|Deliver| L[👤 Customer Receives]
    L -->|Invoice| M[💰 Finance: Create Invoice]
    M -->|Payment| N[💰 Finance: Record Payment]
  
```

Journey 2: Purchase to Production

```

flowchart LR
    A[🏭 Production Needs Materials] -->|Purchase Request| B[🏢 Procurement]
    B -->|Check Warehouse| C{In Stock?}
    C -->|Yes| D[📦 Dispense from Warehouse]
    C -->|No| E[Create Purchase Order]
    E -->|Send to Supplier| F[👤 Supplier Ships]
    F -->|Goods Arrive| G[🔍 QC: Incoming Inspection]
    G -->|Passed| H[📦 Add to Inventory]
    G -->|Failed| I[Return to Supplier]
    H -->|Consume| J[🏭 Production: Material Consumption]
    J -->|Process| K[🏭 Manufacturing Stages]
    K -->|Batch Ready| L[🔍 QC: Finished Product]
    L -->|Passed| M[📦 Finished Goods Inventory]
  
```

Journey 3: Employee Lifecycle

```

flowchart LR
    A[📄 Recruitment/ATS] -->|AI CV Score| B{Hired?}
    B -->|Yes| C[👤 Create Employee]
    C -->|Auto Permissions| D[🔑 Role + Permissions]
    D -->|Daily| E[📅 Attendance + Leave + Overtime]
    E -->|Monthly| F[💰 Payroll Calculation]
    F -->|Payment| G[💰 Finance: Salary Payment]
    G -->|If Injured| H[🏥 Clinic Visit]
    H -->|Performance| I[📊 Performance Review]
    I -->|Assigned| J[🖨 Assets + Equipment]
    J -->|Eventually| K[🚪 Offboarding]
    K -->|Clearances| L[IT + HR + Finance Clear]
    L -->|Complete| M[Account Deactivated]
  
```

PART 26 — MASTER ERD

```

erDiagram
    COMPANIES ||--o{ BRANCHES : "has"
    BRANCHES ||--o{ USERS : "employs"
    USERS ||--o{ ATTENDANCE : "tracks"
    USERS ||--o{ LEAVE_REQUESTS : "submits"
    USERS ||--o{ PAYROLL : "paid"
    USERS ||--o{ LOANS : "borrows"

    SUPPLIERS ||--o{ PURCHASE_ORDERS : "supplies"
    SUPPLIERS ||--o{ SUPPLIER_PERFORMANCE : "evaluated"

    PURCHASE_REQUESTS }o--|| INVENTORY_ITEMS : "for"
    INVENTORY_ITEMS ||--o{ INVENTORY_TRANSACTIONS : "moved"
    INVENTORY_ITEMS ||--o{ BOM_ITEMS : "used_in"

    PRODUCTS ||--o{ BOM : "has_recipe"
    BOM ||--|{ BOM_ITEMS : "contains"
    PRODUCTION_ORDERS ||--o{ PRODUCTION_STAGE_LOGS : "stages"
    PRODUCTION_ORDERS ||--o{ PRODUCTION_BATCHES : "produces"
    PRODUCTION_BATCHES }o--o| QC_INSPECTIONS : "inspected"

    CLIENTS ||--o{ SALES_ORDERS : "places"
    PRODUCTS ||--o{ QUOTE_REQUESTS : "quoted"
    PRODUCT_CATEGORIES ||--o{ PRODUCTS : "categorizes"

    FLEET_VEHICLES ||--o{ FLEET_TRIPS : "used_in"
    FLEET_INTERNAL_DRIVERS ||--o{ FLEET_TRIPS : "drives"

    MACHINES ||--o{ MAINTENANCE_LOGS : "maintained"
    MAINTENANCE_COMPANIES ||--o{ MAINTENANCE_VISITS : "performs"
    EQUIPMENT_CATEGORIES ||--o{ EQUIPMENT : "groups"
    EQUIPMENT ||--o{ EQUIPMENT_RENTALS : "rented"

    USERS ||--o{ CLINIC_VISITS : "visits"
    CLINIC_MEDICATIONS ||--o{ CLINIC_DISPENSE_LOGS : "dispensed"

    FINANCE_INVOICES ||--o{ FINANCE_COLLECTIONS : "paid"
    FINANCE_JOURNAL_ENTRIES ||--|{ FINANCE_JOURNAL_LINES : "contains"
    CHART_OF_ACCOUNTS ||--o{ CHART_OF_ACCOUNTS : "parent"

    USERS ||--o{ ENGINEERING_PROJECTS : "manages"
    ENGINEERING_PROJECTS ||--o{ ENGINEERING_DRAWINGS : "has"

    APPROVAL_WORKFLOWS ||--|{ APPROVAL_STEPS : "has"
    APPROVAL_WORKFLOWS ||--o{ APPROVAL_REQUESTS : "processes"

    TRAINING_COURSES ||--o{ TRAINING_ENROLLMENTS : "enrolls"
    USERS ||--o{ PERFORMANCE_REVIEWS : "reviewed"

```

PART 27 — MASTER PROCESS FLOW

```

flowchart TD
    subgraph "🔒 Authentication"
        LOGIN[Login] --> PERMS[Permission Check]
    end

    subgraph "👥 HR Operations"
        PERMS --> ATTEND[Attendance]
        PERMS --> LEAVES[Leave Management]
        PERMS --> RECRUIT[Recruitment/ATS]
    end

    subgraph "💰 Compensation"
        ATTEND --> PAYROLL[Payroll Calc]
        LEAVES --> PAYROLL
        PAYROLL --> PAYMENT[Salary Payment]
    end

    subgraph "📦 Procurement"
        PR[Purchase Request] --> WCHECK{Stock Check}
        WCHECK -->|Available| DISP[Dispense]
        WCHECK -->|Unavailable| PO[Purchase Order]
        PO --> RECV[Goods Receipt]
        RECV --> QCI[QC: Incoming]
    end

    subgraph "📦 Inventory"
        QCI -->|Pass| STOCK_IN[Stock In]
        DISP --> STOCK_OUT[Stock Out]
    end

    subgraph "🏭 Production"
        STOCK_OUT --> CONSUME[Material Consumption]
        CONSUME --> STAGES[Manufacturing Stages]
        STAGES --> BATCH[Create Batch]
    end

    subgraph "🔍 Quality"
        BATCH --> QCF[QC: Finished Product]
        QCF -->|Pass| FG[Finished Goods]
    end

    subgraph "🏪 Sales"
        QUOTE[Quote Request] --> SO[Sales Order]
        SO --> FG
    end

    subgraph "🚚 Fleet"
        SO --> TRIP[Schedule Trip]
        TRIP --> DELIVER[Delivery]
    end

    subgraph "💵 Finance"
        DELIVER --> INV_S[Sales Invoice]
        PO --> INV_P[Purchase Invoice]
        PAYMENT --> GL[General Ledger]
        INV_S --> GL
    end

```

```
INV_P --> GL
end
```

PART 30 — FINAL VALIDATION REPORT

Validation Summary

Category	Status	Count
Tables Documented	✔ Complete	137+
ERD Diagrams Created	✔ Complete	15 Section ERDs + 1 Master
Process Flows	✔ Complete	25+ Processes
Business Rules	✔ Complete	90+ Rules
CRUD Analysis	✔ Complete	All 15 Modules
Permissions Matrix	✔ Complete	All Modules
Relationships	✔ Complete	All FK Documented
Cross-Module Flows	✔ Complete	3 Data Journeys

Issues & Recommendations

#	Issue	Impact	Recommendation
1	All RLS policies are OPEN (<code>USING (true)</code>)	⚠ Security	Implement proper RLS policies based on user_id and role
2	Legal module is minimal (1 table)	💠 Feature Gap	Add case categories, documents, deadlines, costs
3	No production_orders CREATE TABLE found in read SQL files	💠 Missing DDL	Confirm table exists or create it — it's referenced by FK
4	<code>users.department</code> and <code>users.role</code> CHECK constraints were DROPPED	⚠ Data Integrity	Use a separate <code>departments</code> table with FK instead of varchar
5	Custom auth (not Supabase Auth) stores password_hash in <code>users</code>	⚠ Security	Consider using Supabase Auth or adding bcrypt/argon2 verification
6	<code>late_deductions</code> uses employee_name TEXT, not FK	💠 Data Integrity	Add employee_id UUID FK reference
7	<code>fleet_vehicles</code> uses Arabic status values	💠 Consistency	Use English status codes with Arabic translations in UI
8	No explicit <code>indexes</code> on many FK columns	💠 Performance	Add indexes on all FK columns for join performance
9	Missing <code>spare_parts_requests</code> in detailed analysis	💠 Completeness	Table exists but needs full feature integration
10	No <code>driver_costs</code> table found (referenced in codebase)	💠 Missing DDL	Create or confirm table in production DB

🟢 Suggestions for Future Addition (المش موجود — اقتراحات للإضافة لاحقاً)

#	Suggested Feature	Module	Reason
1	Employee Self-Service Portal (mobile)	HR	Allow employees to check-in/out, view salary, submit requests via mobile

#	Suggested Feature	Module	Reason
2	Shift Scheduling System	HR	Automated shift planning with swap management
3	Production Planning & Scheduling	Production	MRP (Material Requirements Planning) integration
4	Customer Portal (full)	Sales	Allow customers to track orders, view invoices, pay online
5	Supplier Portal (enhanced)	Procurement	Allow suppliers to submit quotations, update delivery status
6	Real-time Production Dashboard	Production	Live OEE metrics, machine utilization, waste tracking
7	Automated Inventory Reordering	Inventory	Auto-generate POs when stock hits reorder point
8	Multi-Currency Invoicing	Finance	Issue invoices in USD, EUR, SAR alongside EGP
9	Project Costing	Engineering	Track costs per engineering project
10	Predictive Maintenance	Maintenance	AI-based failure prediction using maintenance history
11	Barcode/RFID Warehouse	Inventory	Barcode scanning for stock movements
12	Advanced Legal Module	Legal	Case management, documents, lawyer fees, court dates
13	Employee Loan History (credit score)	Payroll	Track employee borrowing patterns
14	Safety Incident Board	Nursing/HSE	Dashboard tracking factory-wide safety metrics
15	API Gateway	IT	REST API for third-party integrations

=====
 دليل الأقسام — Department Guide

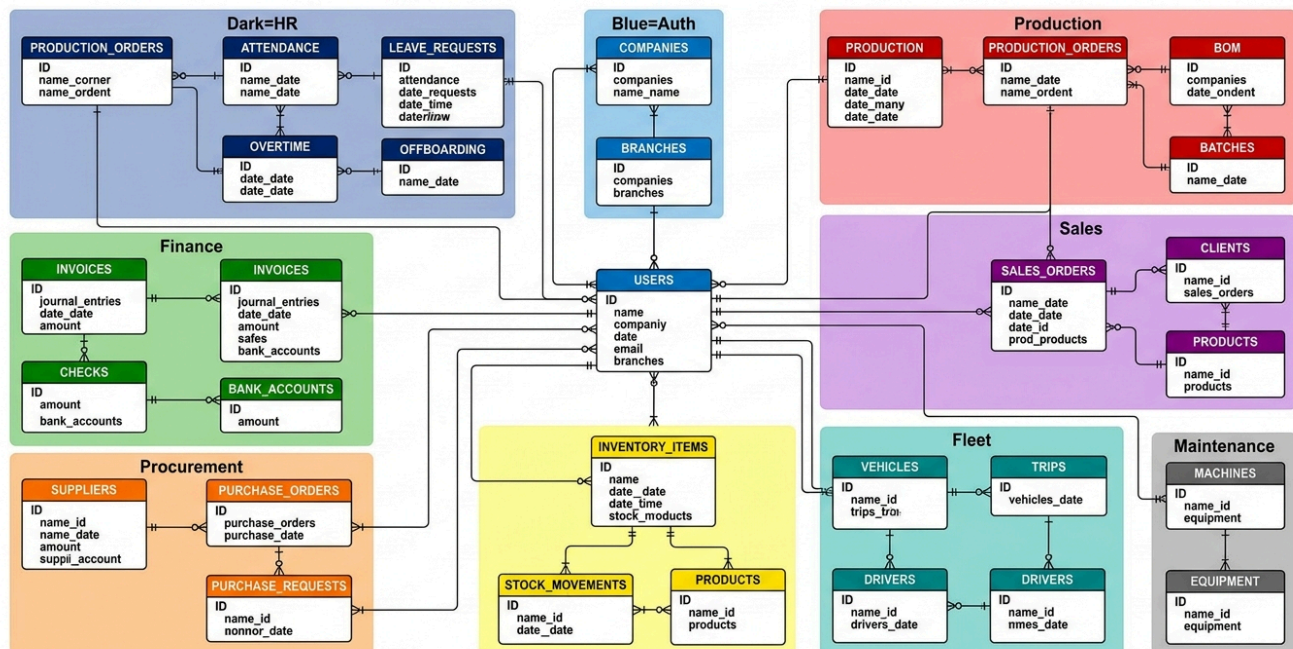
Ninja Factory ERP — All 16 Departments

كل قسم بالشاشات والدورات بتاعته — ليه الصفحات دي في القسم ده

ERD Images

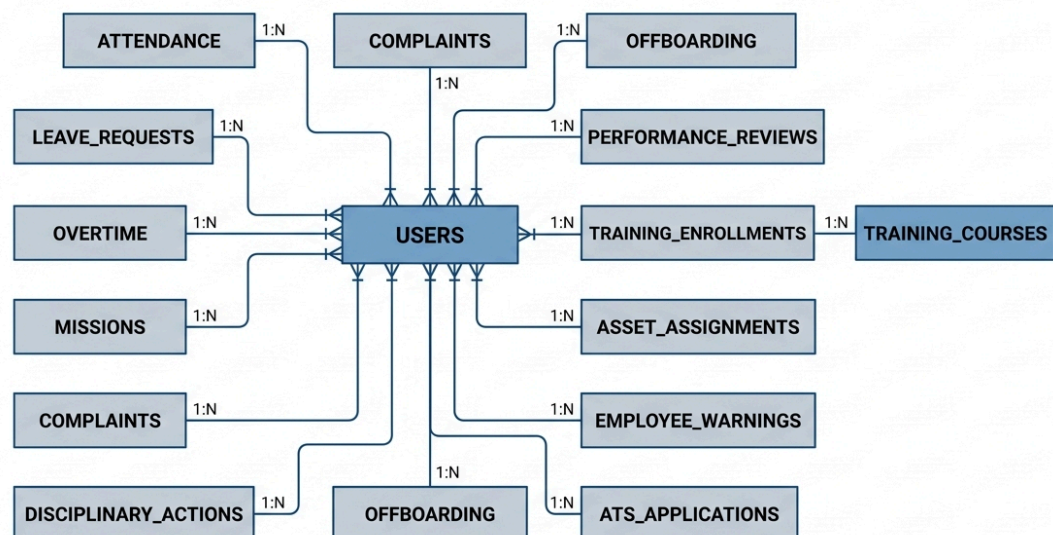
Master System ERD

Ninja Factory ERP — Master ERD



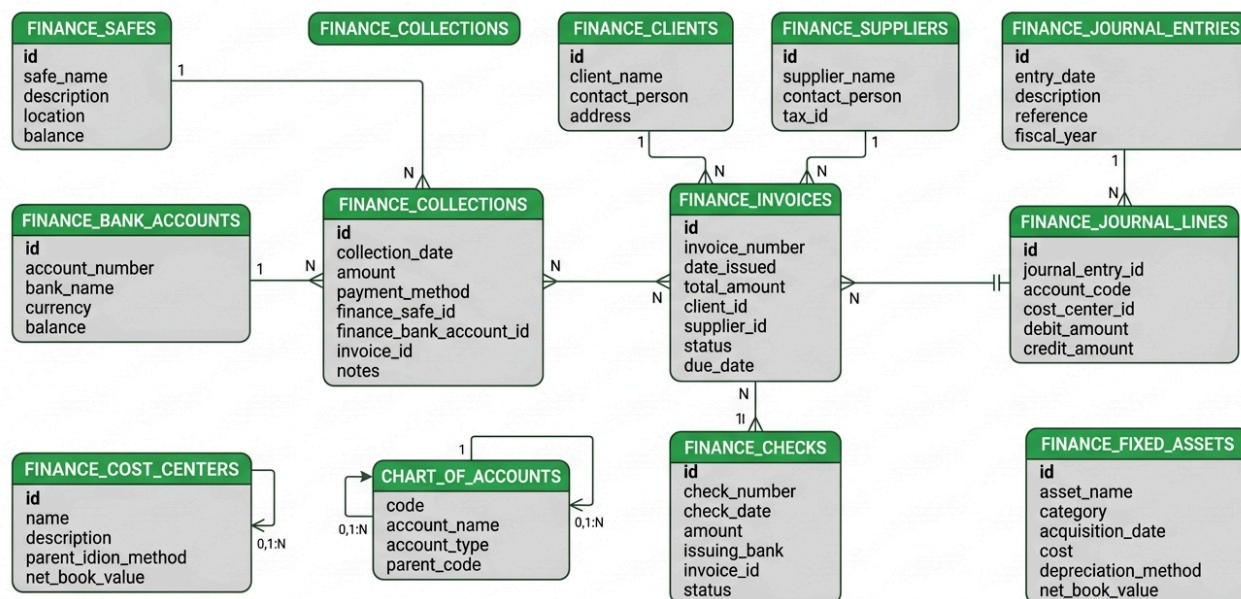
HR & People Management ERD

HR & People Management — ERD



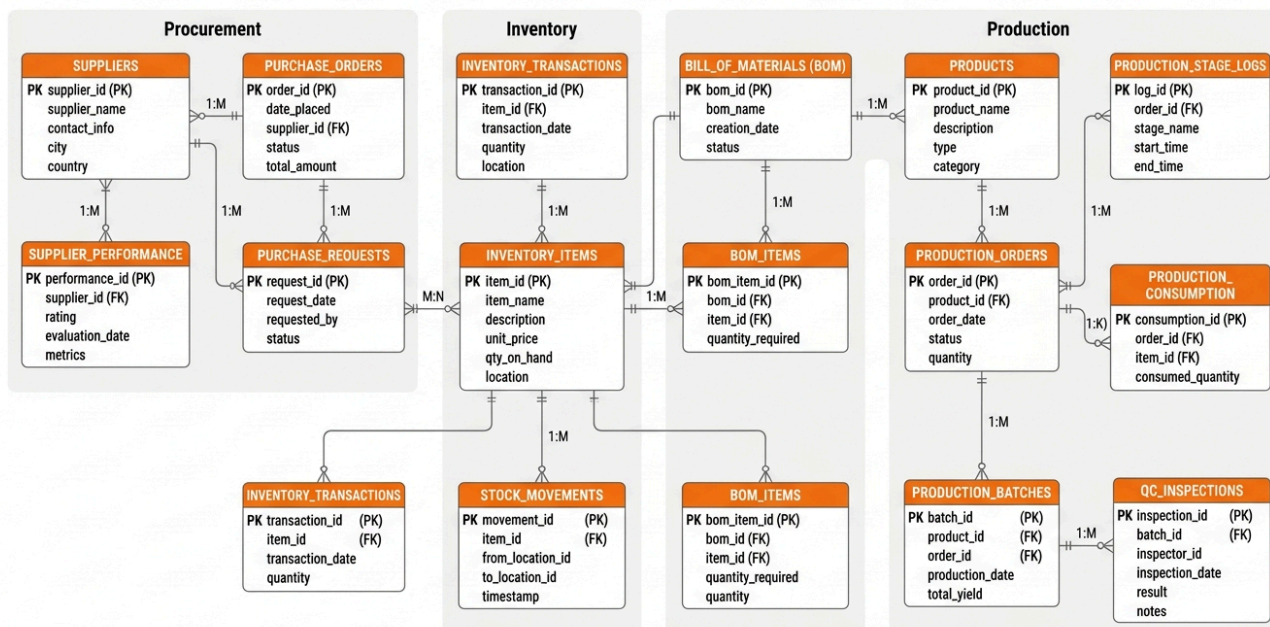
Finance & Accounting ERD

Finance & Accounting — ERD



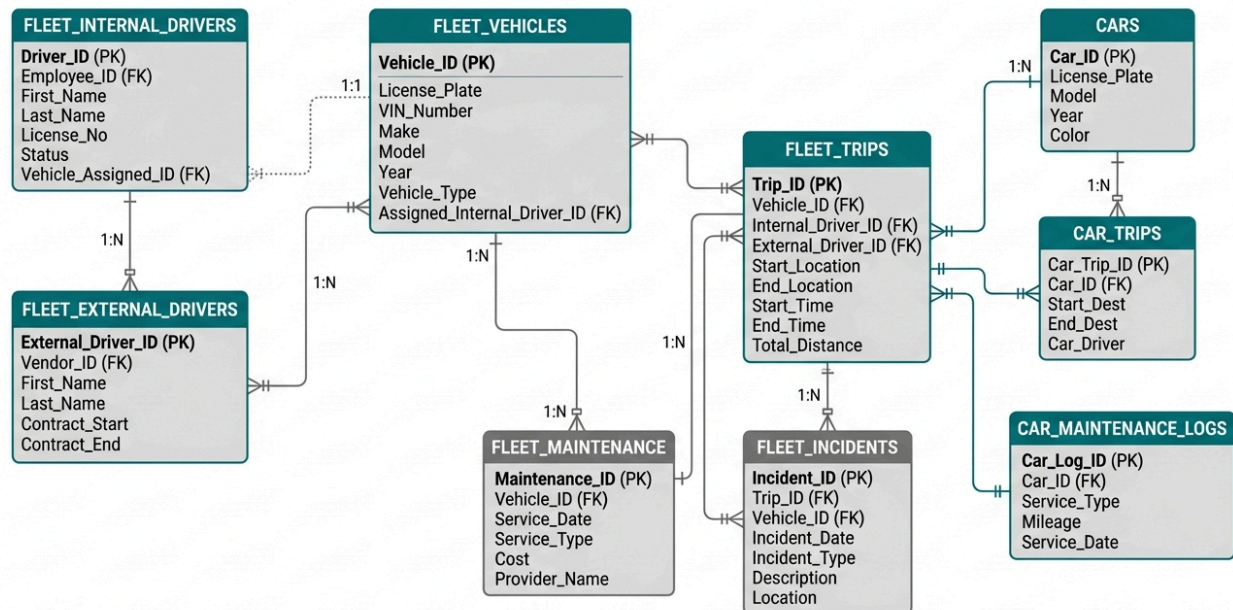
Supply Chain ERD (Procurement → Inventory → Production → QC)

SUPPLY CHAIN — ERD (Procurement → Inventory → Production → QC)



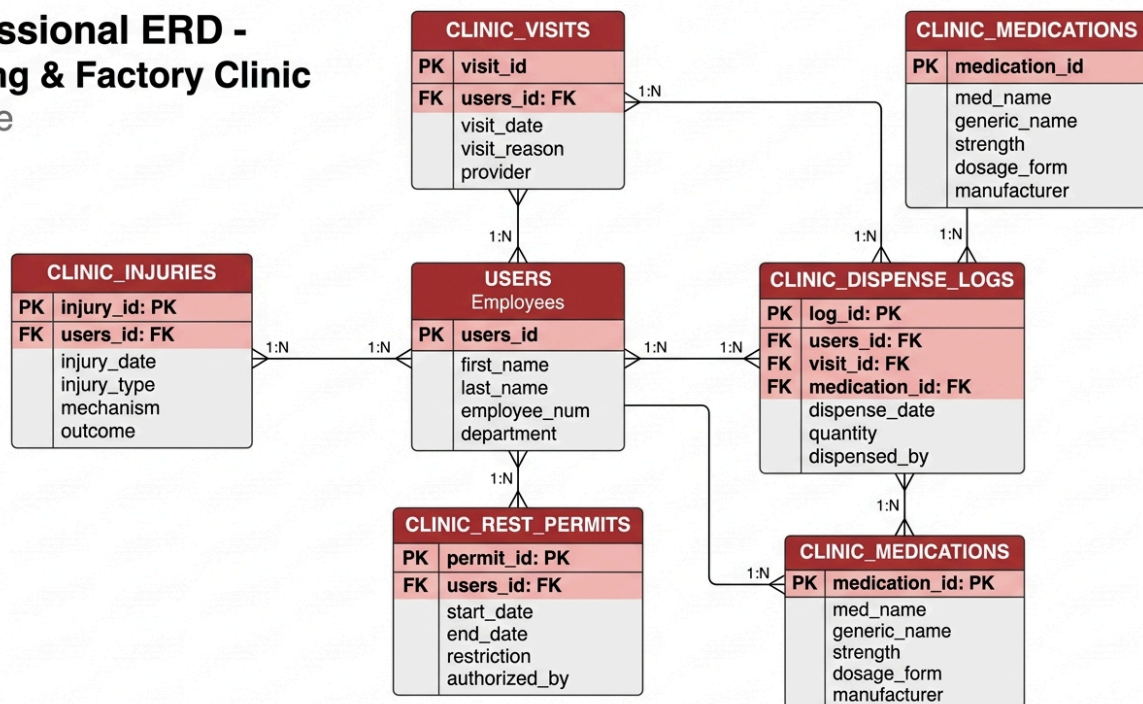
Fleet & Logistics ERD

Fleet & Logistics — ERD



Nursing & Clinic ERD

Professional ERD - Nursing & Factory Clinic Module



1. HR Department — قسم الموارد البشرية

الغرض: إدارة شاملة لدورة حياة الموظف من التعيين حتى الاستقالة

HR Manager, HR Staff: من يعمل هنا

الشاشات ولماذا

#	الشاشة	لماذا موجودة
1	employees	إدارة بيانات جميع الموظفين — إضافة، تعديل، حذف، بحث
2	attendance	متابعة حضور وانصراف جميع الموظفين يومياً
3	leaves	إدارة طلبات الإجازات — مراجعة وموافقة/رفض
4	absence-leave	نظام الغياب المتقدم — تسجيل الغياب غير المبرر وتطبيق الجزاءات
5	shifts	إدارة الورديات — صباحي، مسائي، ليلي، إداري
6	overtime	إدارة ساعات العمل الإضافي والموافقة عليها
7	all-delays	عرض جميع حالات التأخير مع حساب الخصومات
8	all-missions	إدارة المأموريات الخارجية لجميع الموظفين
9	employee-warnings	إصدار الإنذارات والجزاءات التأديبية
10	asset-assignment	تسليم وتتبع العهد (لابتوب، موبايل، مفاتيح، زي)
11	payroll	حساب وصرف المرتبات الشهرية
12	payroll-funding	تمويل المرتبات (موافقة صاحب المصنع)
13	hr-adjustments	تعديلات المرتبات (بدلات، حوافز، خصومات)
14	recruitment	إدارة التوظيف — إعلانات الوظائف
15	hr-ats	للسير الذاتية AI نظام تتبع المتقدمين مع تقييم
16	loans	إدارة سلف الموظفين
17	expenses	إدارة مصروفات الموظفين
18	complaints	إدارة شكاوى الموظفين (سرية أو معلنة)
19	medical-requests	طلبات العلاج الطبي والصرف
20	offboarding	إجراءات إنهاء الخدمة والمخالفات
21	training	إدارة الدورات التدريبية
22	performance-reviews	تقييم أداء الموظفين الدوري
23	hr-qr-generator	للحضور QR توليد أكواد
24	friday-work	طلبات العمل يوم الجمعة
25	documents	إدارة مستندات الموظفين

دورات القسم (Cycles)

الدورة 1: دورة الإجازة

يوافق/يرفض → المالك يوافق/يرفض → خصم من رصيد الإجازات HR → موظف يقدم طلب → مدير الصالة يوافق/يرفض

الدورة 2: دورة الغياب والجزاءات

يمكن يعدل → 3 أيام متتالية → إنذار أول → إنذار نهائي → فصل HR → غياب بدون إذن → خصم يومين تلقائي

الدورة 3: دورة التوظيف

فرز → مقابلة → عرض → تعيين → إنشاء حساب → AI تقييم → CVs إعلان وظيفة → استلام

الدورة 4: دورة إنهاء الخدمة

إغلاق الحساب → HR مخالصة مالية → مخالصة → IT بدء الإجراء → مخالصة

2. قسم المالية — Finance Department

الغرض: إدارة الحسابات والخزينة والمراجعة المالية

من يعمل هنا CFO, Chief Accountant, Accountant

الشاشات

#	الشاشة	لماذا موجودة
1	chart-of-accounts	شجرة الحسابات — تصنيف جميع الحسابات المالية
2	financial-reports	التقارير المالية — ميزان المراجعة، قائمة الدخل
3	petty-cash	إدارة الخزينة والبنوك والعهد
4	driver-payments	مدفوعات السائقين والنقل

دورات القسم

الدورة 1: دورة الفاتورة

إنشاء فاتورة → غير مسددة → تحصيل جزئي → تحصيل كامل → مسددة → قيد يومية

الدورة 2: دورة الشيكات

استلام شيك → قيد انتظار → إيداع بالبنك → تحت التحصيل → محصل / مرتد

الدورة 3: دورة قيود اليومية

إنشاء قيد (مسودة) → مراجعة → اعتماد → ترحيل → تحديث الأرصدة

الدورة 4: دورة الإقفال

إقفال يومي → إقفال شهري → إقفال سنوي → قفل الفترة (لا تعديل)

3. قسم الإنتاج — Production Department

الغرض: إدارة عمليات التصنيع من المواد الخام حتى المنتج النهائي

من يعمل هنا: Production Manager, Planning Manager, Line Workers

الشاشات

#	الشاشة	لماذا موجودة
1	production	إدارة أوامر الإنتاج
2	erp-products	(BOM) إدارة المنتجات ووصفات التصنيع

دورات القسم

الدورة 1: دورة أمر الإنتاج

فحص جودة → منتج نهائي → المخزن → (Batch) طلب خامات من المخزن → استهلاك الخامات → مراحل التصنيع → إنشاء دفعة → BOM + إنشاء أمر إنتاج → اختيار المنتج

(Traceability) الدورة 2: دورة التتبع

خامات مستخدمة → مورد الخامة → BOM → دفعة → أمر إنتاج

4. قسم المخازن — Warehouse Department

الغرض: إدارة المخزون وحركات الأصناف

من يعمل هنا: Warehouse Manager, Storekeepers

الشاشات

#	الشاشة	لماذا موجودة
1	inventory	إدارة جميع أصناف المخزون (خامات، قطع غيار، مستلزمات)
2	purchase-requests	طلبات الشراء من الأقسام — الفحص والصرف

دورات القسم

الدورة 1: دورة حركة المخزون

وارد (من مشتريات) → تخزين → صادر (لإنتاج/أقسام) → تحويل بين مخازن → تسوية

الدورة 2: دورة طلب الشراء من المخزن

قسم يطلب صنف → المخزن يفحص → متوفر: صرف مباشر → غير متوفر: تحويل للمشتريات

5. قسم المشتريات — Procurement Department

الغرض: شراء المواد الخام والمستلزمات وإدارة علاقات الموردين

من يعمل هنا: Procurement Manager, Procurement Specialist

الشاشات

#	الشاشة	لماذا موجودة
1	purchase-requests	مراجعة طلبات الشراء من الأقسام
2	erp-suppliers	إدارة بيانات الموردين
3	supplier-portal	بوابة المورد الخارجي لتتبع الطلبات
4	supplier-performance	تقييم أداء الموردين بالذكاء الاصطناعي

دورات القسم

الدورة 1: دورة الشراء

طلب شراء → موافقة المشتريات → طلب عروض أسعار → إنشاء أمر شراء → إرسال للمورد → استلام البضاعة → فحص الجودة → إدخال المخزن → تسوية مالية

6. قسم الجودة — Quality Department

الغرض: ضمان جودة المواد الواردة والمنتجات النهائية

من يعمل هنا: Quality Manager, QC Inspector

الشاشات

#	الشاشة	لماذا موجودة
1	quality	إدارة فحوصات الجودة

دورات القسم

الدورة 1: فحص المواد الواردة

نجاح: إدخال مخزن → فشل: إرجاع للمورد → QC مادة خام تصل → فحص

الدورة 2: فحص المنتج النهائي

نجاح: منتج نهائي → فشل: إعادة تصنيع / خردة → QC دفعة إنتاج جاهزة → فحص

7. قسم المبيعات — Sales Department

الغرض: إدارة المبيعات والعملاء وكتالوج المنتجات

من يعمل هنا: Sales Manager, Sales Staff

الشاشات

#	الشاشة	لماذا موجودة
1	erp-sales	إدارة أوامر البيع
2	erp-products	إدارة كتالوج المنتجات (عام ومخفي)
3	customer-requests	طلبات تسجيل العملاء الجدد وعروض الأسعار

دورات القسم

الدورة 1: دورة البيع

إرسال عرض → موافقة العميل → إنشاء أمر بيع → إنتاج/صرف من المخزن → توصيل → فاتورة → تحصيل → Sales عميل يطلب عرض سعر → مراجعة

الدورة 2: تسجيل عميل جديد

يراجع → موافقة → إنشاء حساب عميل Sales → زائر يملأ نموذج تسجيل

8. قسم الهندسة — Engineering Department

الغرض: إدارة المشاريع الهندسية والرسومات الفنية

من يعمل هنا: Engineering Manager, Engineer, Technical Office

الشاشات

#	الشاشة	لماذا موجودة
1	engineering	إدارة المشاريع الهندسية والرسومات

دورات القسم

الدورة 1: دورة المشروع الهندسي

تخطيط → تصميم → رفع رسومات → اعتماد الرسومات → تنفيذ → إشراف → انتهاء → اعتمادات الأقسام

9. قسم الصيانة — Maintenance Department

الغرض: صيانة الماكينات والمعدات وإدارة شركات الصيانة الخارجية

من يعمل هنا : Maintenance Manager, Technicians

الشاشات

#	الشاشة	لماذا موجودة
1	maintenance	إدارة الماكينات وسجلات الصيانة
2	equipment	إدارة المعدات والإيجارات

دورات القسم

الدورة 1: دورة الصيانة الداخلية

عطل/موعد صيانة → تعيين فني → إصلاح → تسجيل السجل → تحديث حالة الماكينة

الدورة 2: دورة الصيانة الخارجية

طلب صيانة → تعيين شركة خارجية → زيارة فنية → تنفيذ الأعمال → تسجيل التكاليف → فاتورة → دفع

الدورة 3: دورة إيجار المعدات

حاجة لمعدة → إنشاء عقد إيجار → تفعيل → استخدام → إنتهاء/تجديد → إرجاع

10. Fleet & Logistics — قسم الأسطول واللوجستيات

الغرض: إدارة السيارات والسائقين والرحلات

من يعمل هنا : Logistics Manager, Internal Drivers

الشاشات

#	الشاشة	لماذا موجودة
1	fleet	إدارة السيارات والسائقين والرحلات

دورات القسم

الدورة 1: دورة الرحلة

جدولة رحلة → تعيين سائق + سيارة → تحميل → انطلاق → في الطريق → توصيل → تسجيل المسافة والوقود

الدورة 2: دورة صيانة الأسطول

تنبيه صيانة (عداد الكيلومترات) → طلب صيانة → تنفيذ → تسجيل التكلفة → تحديث حالة السيارة

11. Nursing & Clinic — قسم التمريض والعيادة

الغرض: إدارة العيادة الطبية للمصنع

من يعمل هنا: Nurse, Nursing Management

الشاشات

#	الشاشة	لماذا موجودة
1	nursing-page	تسجيل زيارات العيادة والإصابات والأدوية
2	nursing-medical-approvals	اعتمادات طبية

دورات القسم

الدورة 1: دورة الزيارة الطبية

موظف يزور العيادة → تسجيل العلامات الحيوية → تشخيص → علاج/صرف دواء → قرار (رجوع للعمل / راحة / تحويل مستشفى)

الدورة 2: دورة إصابة العمل

إصابة → تسجيل الإصابة → إسعافات أولية → تقييم الخطورة → تحويل مستشفى (إذا لزم) → تحقيق السلامة → متابعة

الدورة 3: دورة صيدلية العيادة

استلام أدوية → تخزين → صرف للموظفين → تتبع المخزون → تنبيه نقص → طلب إعادة توريد

12. IT Department — قسم تكنولوجيا المعلومات

الغرض: دعم فني وإدارة النظام والصلاحيات

من يعمل هنا: IT Admin, IT Support

الشاشات

#	الشاشة	لماذا موجودة
1	screen-permissions	إدارة صلاحيات الشاشات والأفعال لكل دور
2	system-settings	إعدادات النظام العامة
3	audit-log	سجل المراجعة — تتبع جميع العمليات
4	login-history	سجل تسجيل الدخول — متابعة أمنية
5	activity-log-page	سجل النشاطات التفصيلي
6	notification-settings	إعدادات الإشعارات
7	ai-mind	المحادثة مع الذكاء الاصطناعي

دورات القسم

الدورة 1: دورة تذكرة الدعم الفني

موظف يفتح تذكرة → تعيين فني → جاري الحل → تم الحل

الدورة 2: دورة إدارة الصلاحيات

مستخدم جديد → تحديد الدور → تطبيق قالب الصلاحيات → تعديل فردي إذا لزم → بث التغييرات فوري

13. قسم التعبئة والتغليف — Packaging Department

الغرض: تعبئة وتغليف المنتجات النهائية

من يعمل هنا: Packaging Workers, Hall Manager

الشاشات

يستخدم الشاشات المشتركة: `dashboard` , `my-attendance` , `my-leaves` , `scan-checkin` , `scan-checkout`

دورات القسم

استلام منتج نهائي من الإنتاج → تعبئة → تغليف → تخزين في مخزن المنتجات النهائية

14. قسم التخطيط — Planning Department

الغرض: تخطيط الإنتاج والجدولة

من يعمل هنا: Planning Manager

الشاشات

KPIs + يستخدم شاشات الإنتاج + التقارير

دورات القسم

تحليل طلبات البيع → تخطيط الإنتاج → جدولة التوريدات → متابعة التنفيذ → تعديل الخطة

15. قسم الصحة والسلامة — Health & Safety Department

الغرض: السلامة المهنية وتتبع الإصابات

من يعمل هنا: HSE Officers

الشاشات

يستخدم شاشات التمريض (إصابات العمل) + التقارير

دورات القسم

تقييم المخاطر → وضع إجراءات وقائية → تسجيل الحوادث → تحقيق → إجراءات تصحيحية

16. الأقسام الإدارية — Administration & Secretariat

الغرض: الشؤون الإدارية والسكرتارية والقانونية

من يعمل هنا: Admin Staff, Legal Advisor

الشاشات

#	الشاشة	لماذا موجودة
1	legal-affairs	تتبع القضايا القانونية
2	announcements	إدارة الإعلانات والتعميمات
3	document-management	إدارة المستندات الرسمية

دورات القسم

دورة القضية القانونية:

تسجيل قضية → تعيين محامي/مسؤول → متابعة → حل → إغلاق

Self-Service الشاشات المشتركة لجميع الموظفين

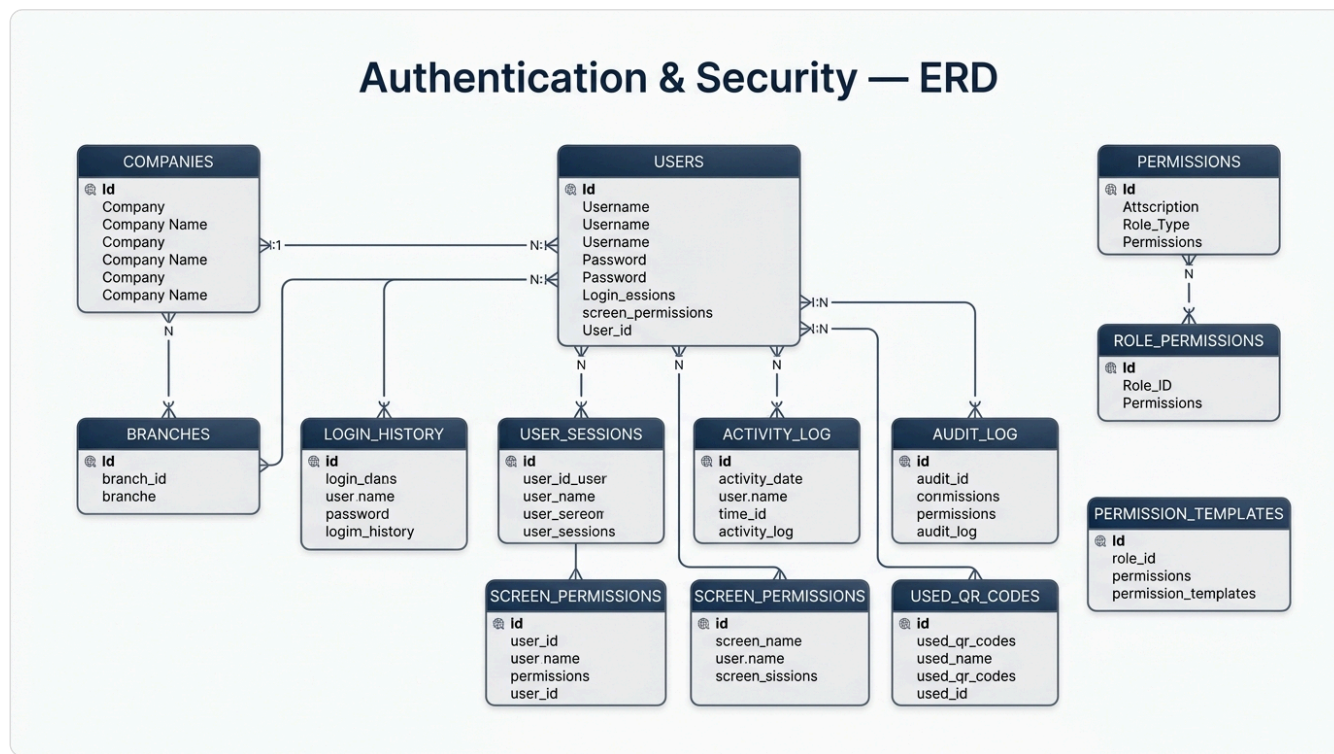
كل موظف في أي قسم يقدر يوصل للشاشات دي:

#	الشاشة	الغرض
1	dashboard	لوحة المعلومات الشخصية
2	hr-personal	بيانات الموظف الشخصية
3	my-attendance	سجل الحضور الشخصي
4	scan-checkin	QR تسجيل الدخول بـ
5	scan-checkout	QR تسجيل الخروج بـ
6	my-leaves	طلبات الإجازات
7	my-salary	كشف المرتب
8	my-overtime	ساعات العمل الإضافي

#	الشاشة	الغرض
9	my-loans	السلف
10	my-medical	طلبات العلاج
11	my-delays	حالات التأخير
12	my-missions	المأموريات
13	my-expenses	المصروفات
14	complaints	تقديم شكوى
15	announcements	عرض الإعلانات
16	internal-chat	المحادثات الداخلية
17	shift-swap	تبديل الورديات
18	calendar	التقويم
19	task-management	إدارة المهام
20	org-directory	دليل المنظمة

ERD Diagrams

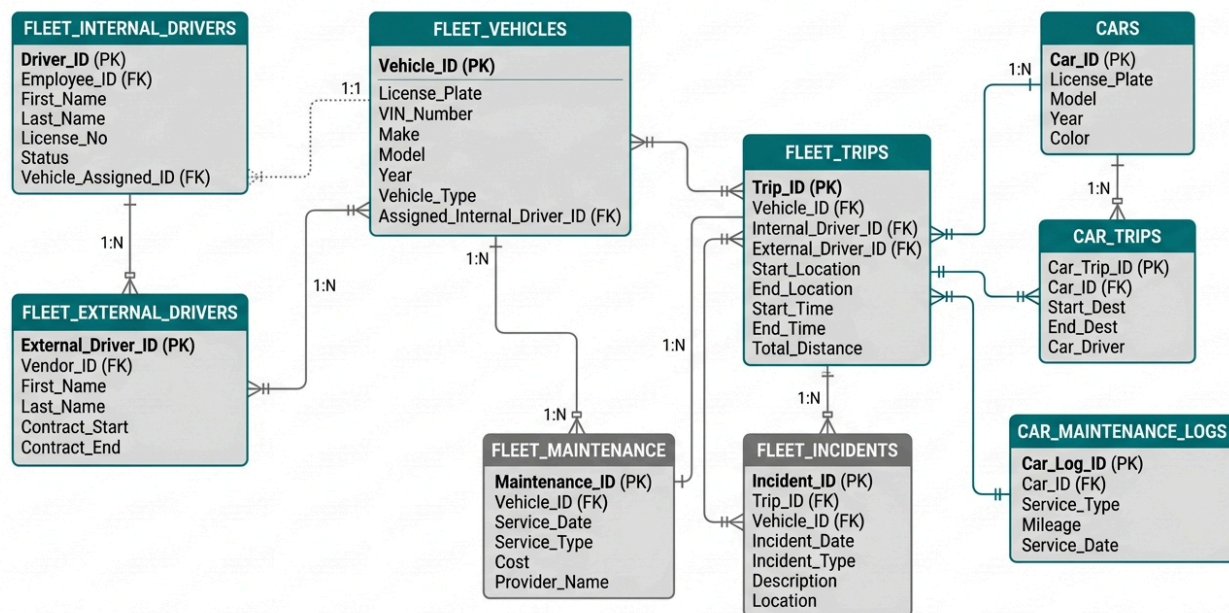
Authentication & Security ERD



Finance & Accounting — ERD

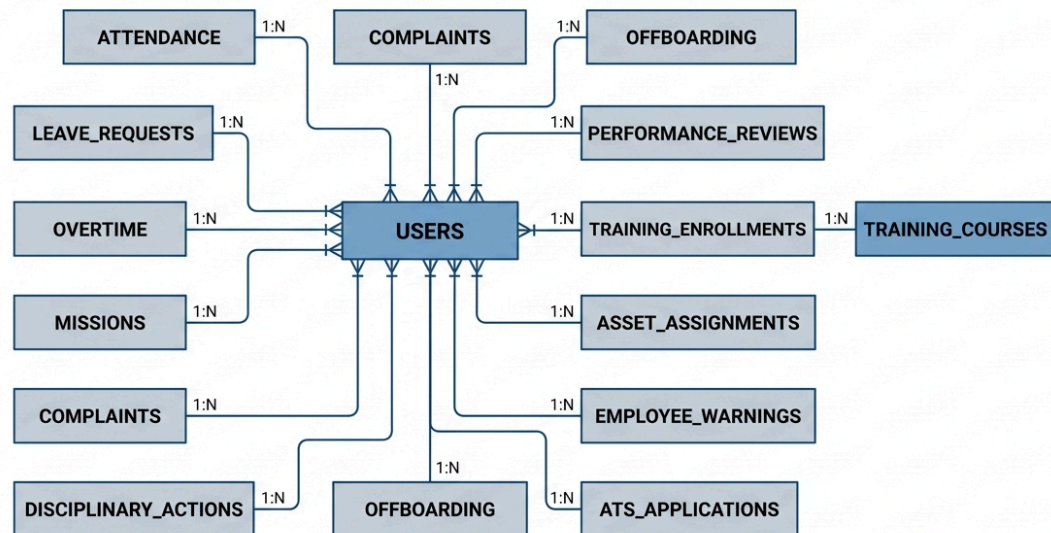


Fleet & Logistics — ERD



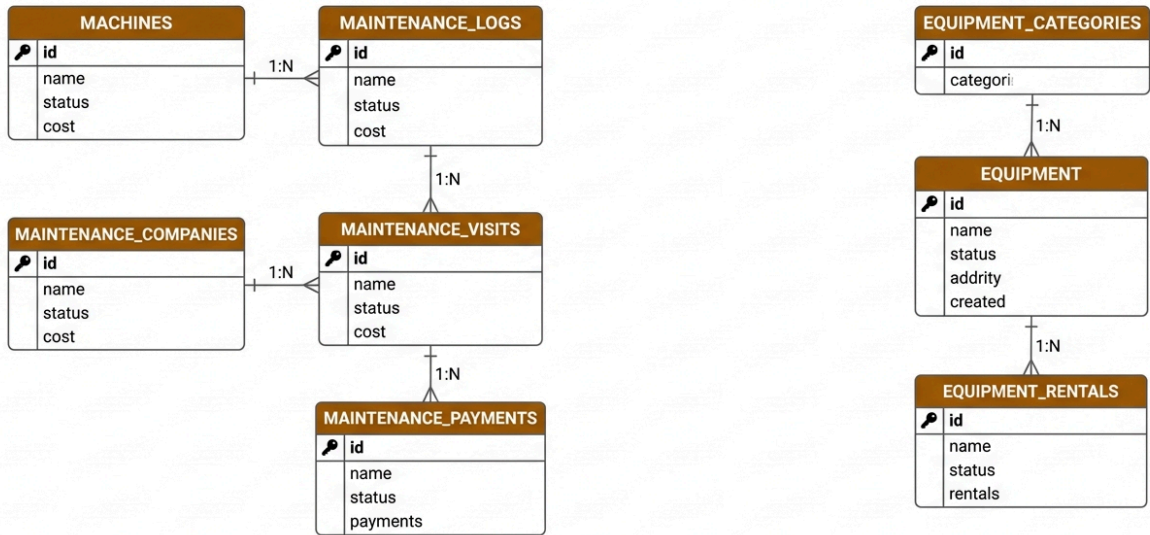
HR & People Management ERD

HR & People Management — ERD

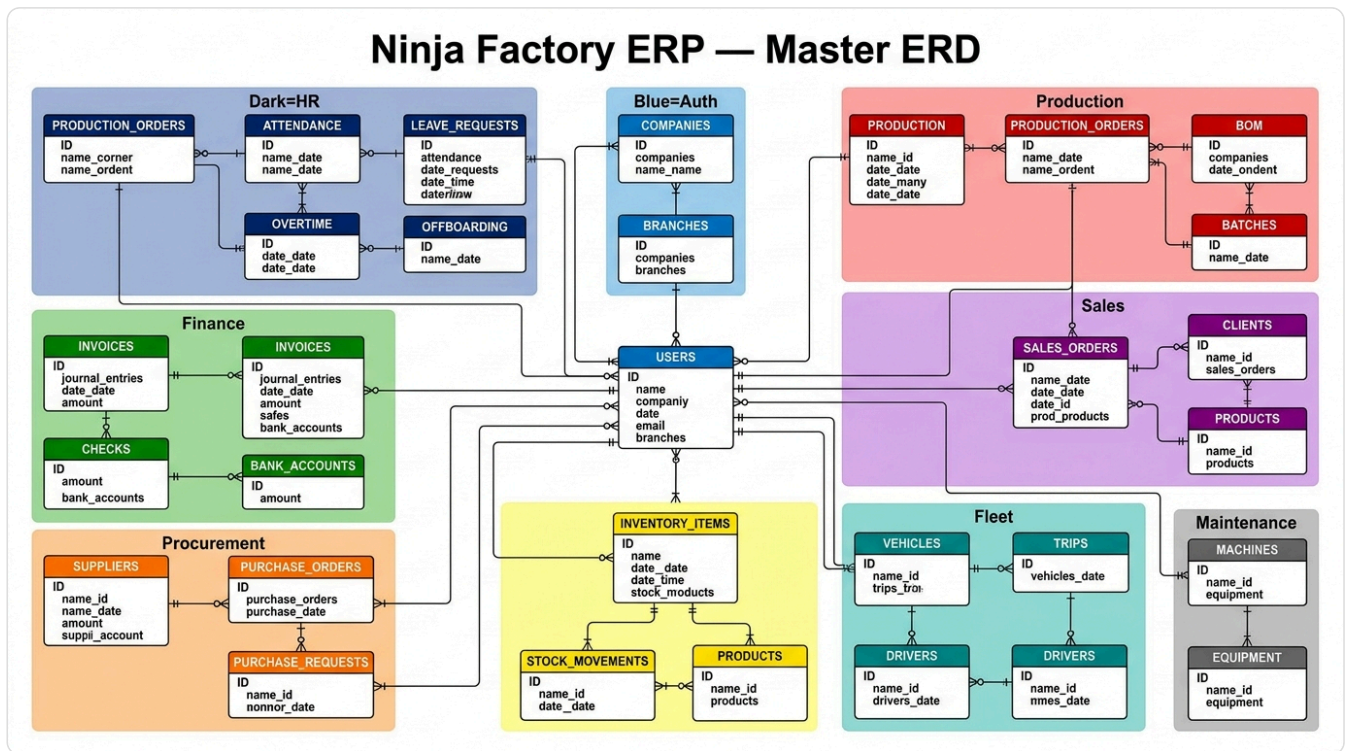


Maintenance & Equipment ERD

Maintenance & Equipment — ERD

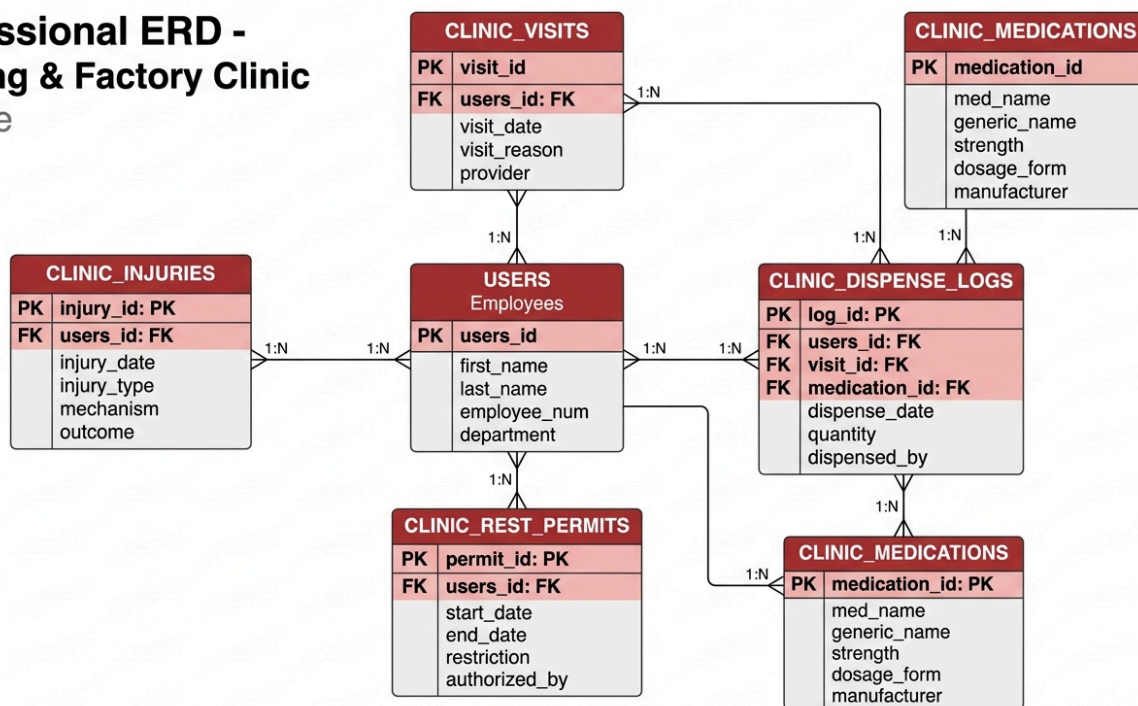


Ninja Factory ERP — Master ERD

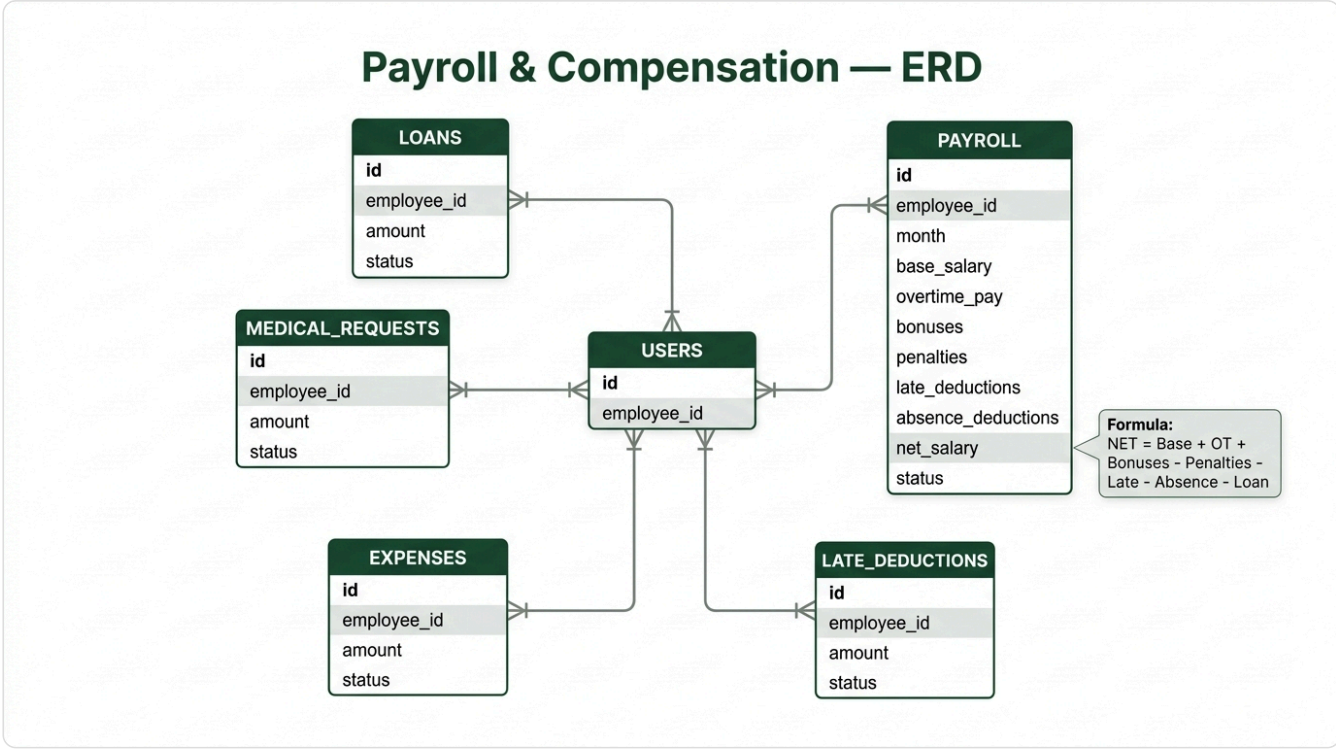


Nursing & Clinic ERD

Professional ERD - Nursing & Factory Clinic Module

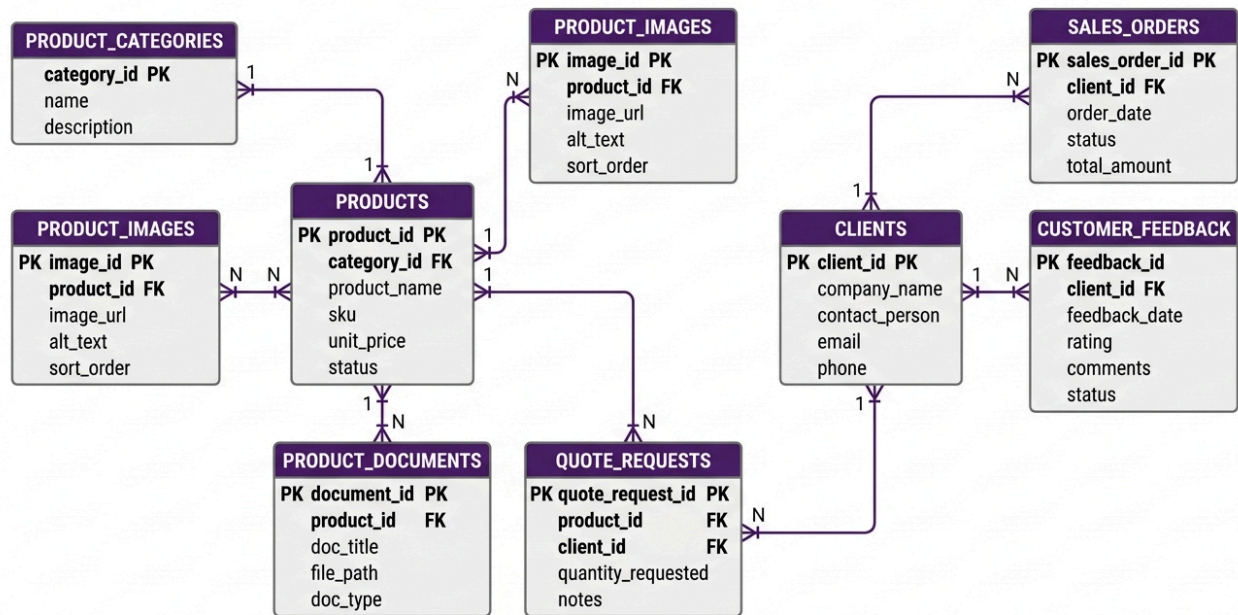


Payroll & Compensation ERD



Sales & CRM ERD

Sales & CRM – ERD



Supply Chain ERD

SUPPLY CHAIN — ERD (Procurement → Inventory → Production → QC)

